MASTER'S DEGREE IN PHARMACEUTICAL BIOTECHNOLOGY

From Genome to Drug – Exploring Bioinformatics & Omics Sciences

Lesson I – Bioinformatics, Omics Sciences & Genomics

Advances in Genomic Analysis Techniques

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Learning Objectives — Part I

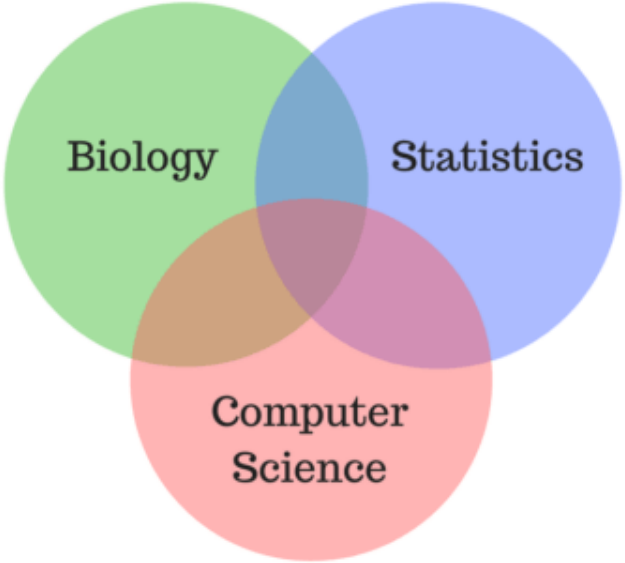
1. Understand the role and scope of Bioinformatics as an interdisciplinary field
2. Know the Central Dogma of Molecular Biology and its relevance to drug discovery
3. Describe the Human Genome Project milestones and their impact on genomics
4. Identify the major OMICS disciplines (Genomics, Transcriptomics, Proteomics, etc.)
5. Explain Next-Generation Sequencing (NGS): principles, technologies, and applications
6. Apply OMICS knowledge to select the correct technique for a clinical scenario (case studies)



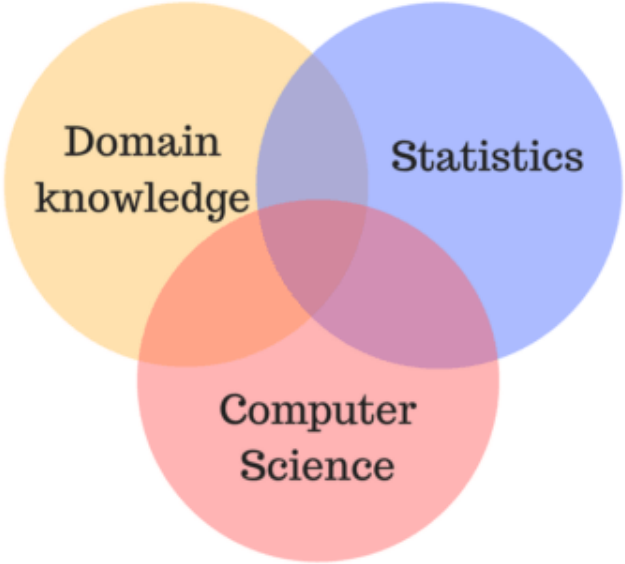
Bioinformatics: Foundations

The interdisciplinary field that bridges biology, computer science, mathematics and statistics to store, analyse and interpret vast amounts of biological data.

Bioinformatics

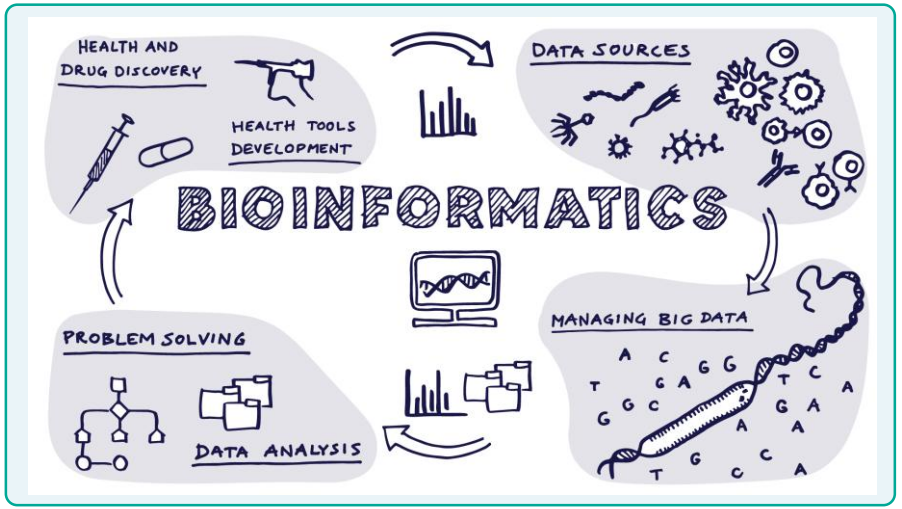


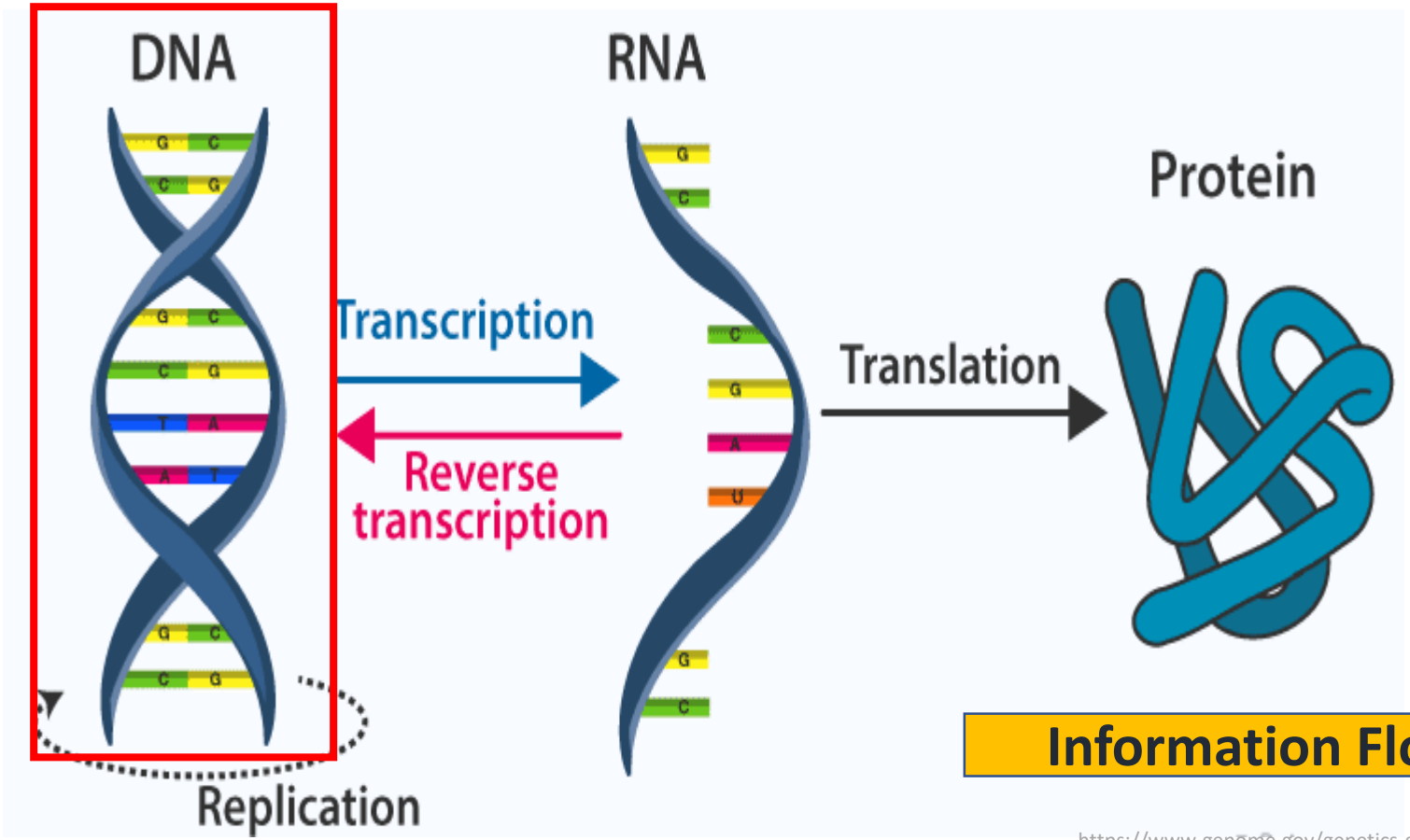
Data Science



What is Bioinformatics?

- **Core mission: convert raw experimental data into meaningful biological knowledge**
- Key tasks include: ***sequence analysis***, ***structural prediction***, ***genome assembly***, ***network modelling***
- Drives research in medicine, **biotechnology**, pharmacology and evolutionary biology
- Emerged from the need to **manage data from sequencing technologies** (especially post-HGP)

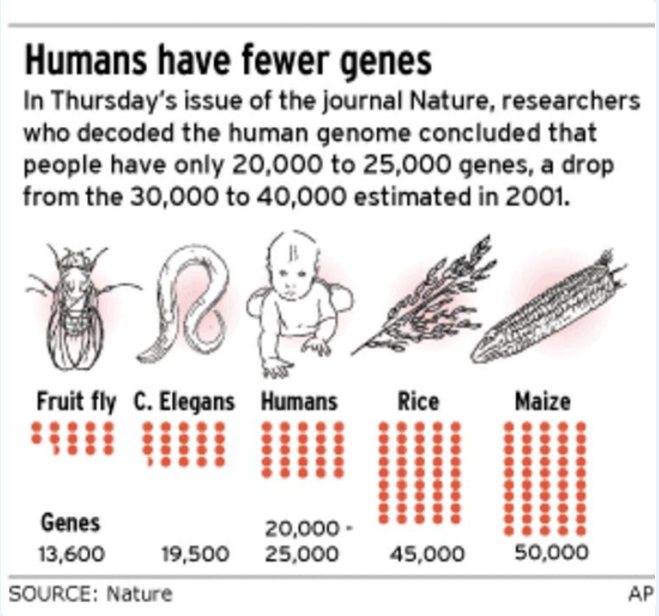




https://www.genome.gov/genetics-glossary/Central-Dogma

Human Genome Project (HGP) — Key Milestones

- **1990** – Official launch; 15-year timeline established
- 1998 – ***Celera Genomics*** enters, accelerating competition & progress
- 2000 – First draft sequence of the human genome announced
- **2001** – Draft published simultaneously in Nature & Science
- 2003 – Official completion (2 years ahead of schedule), coinciding with 50th anniversary of DNA structure discovery
- **Revolutionary finding: ~20,000–22,000 protein-coding genes (far fewer than the expected 100,000)**
- **Key insight: Biological complexity ≠ gene count alone — regulation, epigenetics, and non-coding RNA matter**



Sequencing Cost: From $3 Billion to <$1,000



$3B

Cost — Sanger
(1990–2003)



$1M→$10K

Cost — NGS
(~2007–2015)



<$200

Cost — Modern WGS
(today)



10+ years

Time to sequence
one genome (Sanger)

Moore's Law vs. Sequencing Cost

- Moore's Law predicts computing power doubles every ~2 years (cost halves similarly)
- Sequencing costs initially followed Moore's Law — then dropped FAR faster after NGS arrived (~2007)
- This dramatic cost reduction opened whole-genome sequencing to clinical settings and individual patients
- Implication for drug discovery: large-scale genomic datasets are now accessible and affordable

Central Dogma of Molecular Biology

Proposed by Francis Crick (1958) — describes how genetic information flows inside a cell

DNA

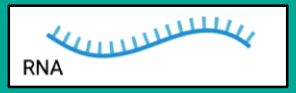
Double-stranded nucleotide sequence



Transcription →

RNA

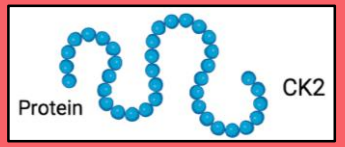
Single-stranded messenger



Translation →

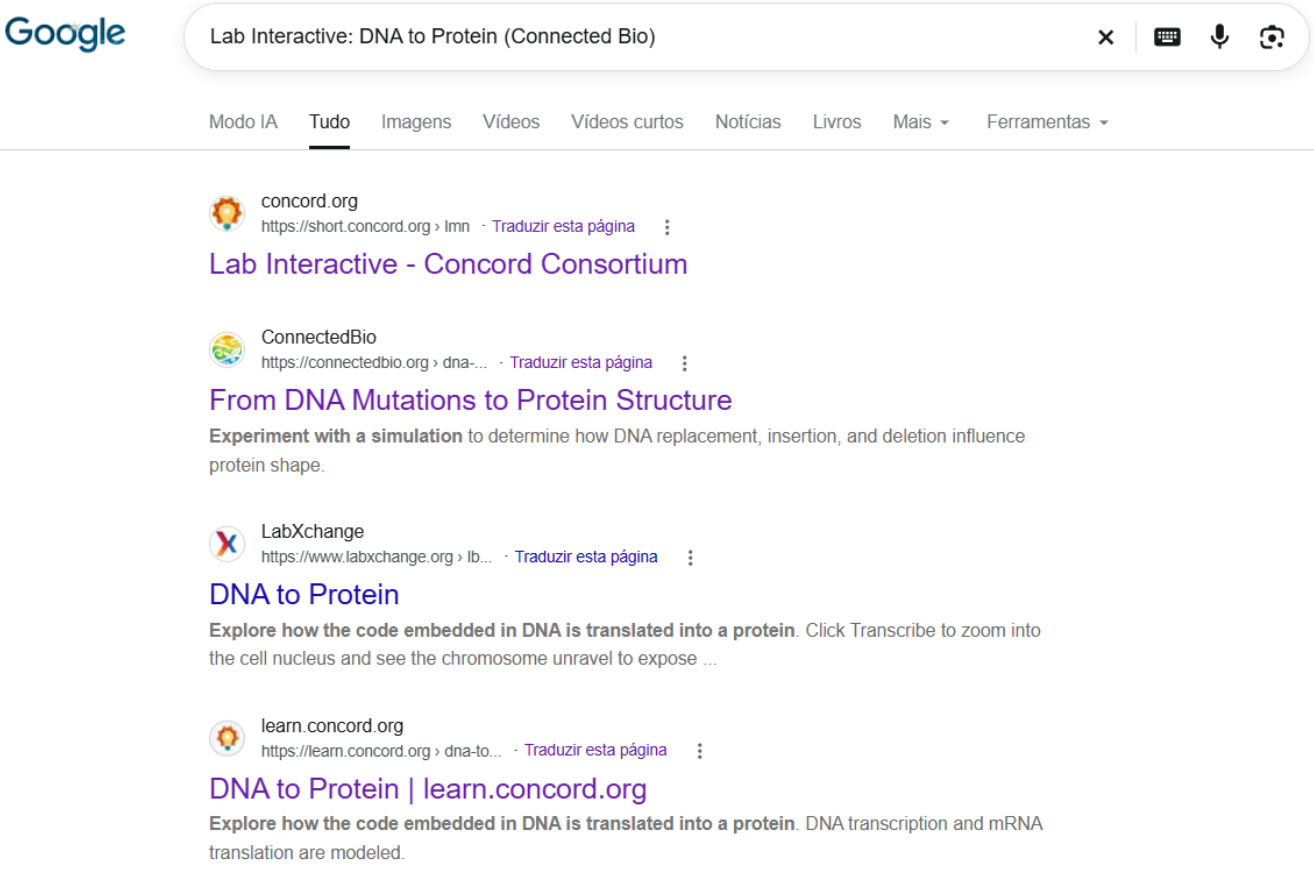
Protein

Amino acid chain
→ functional molecule

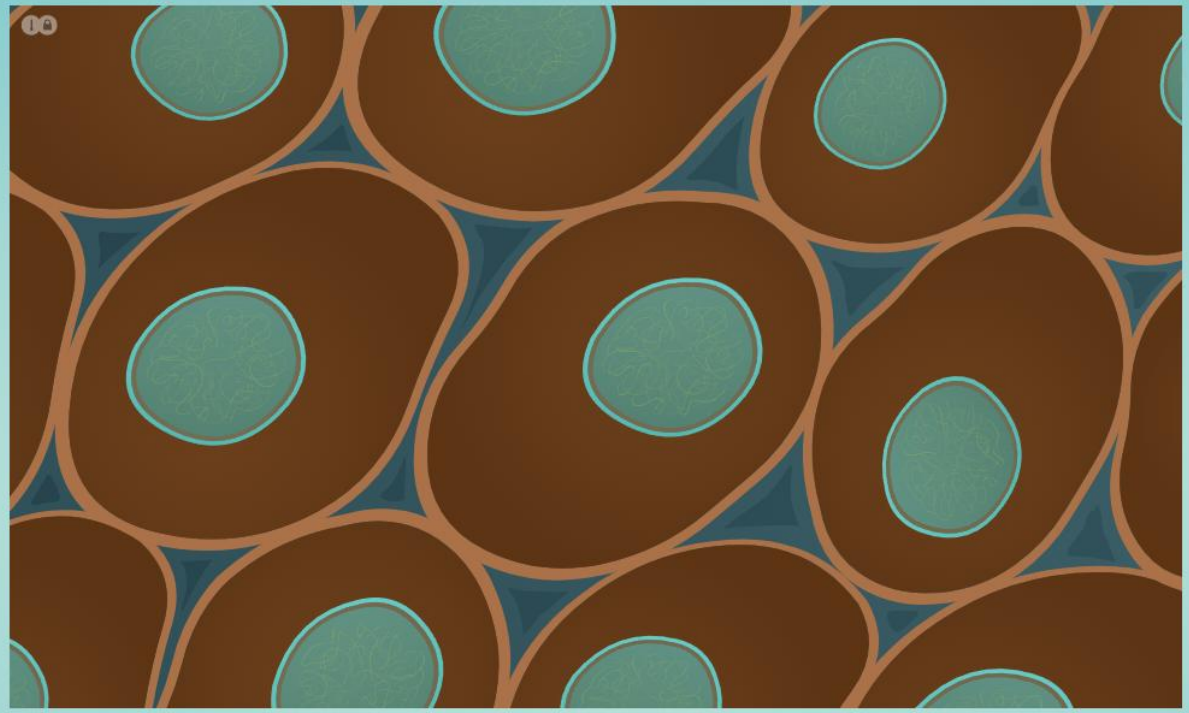


⚠ Exception — Reverse Transcription: **Some retroviruses (e.g., HIV)** use the enzyme **reverse transcriptase** to copy RNA back into DNA. This is exploited in research (RT-PCR) and in antiretroviral drug development.

Lab Interactive - Concord Consortium



Step 1: Enter or Edit a DNA Sequence



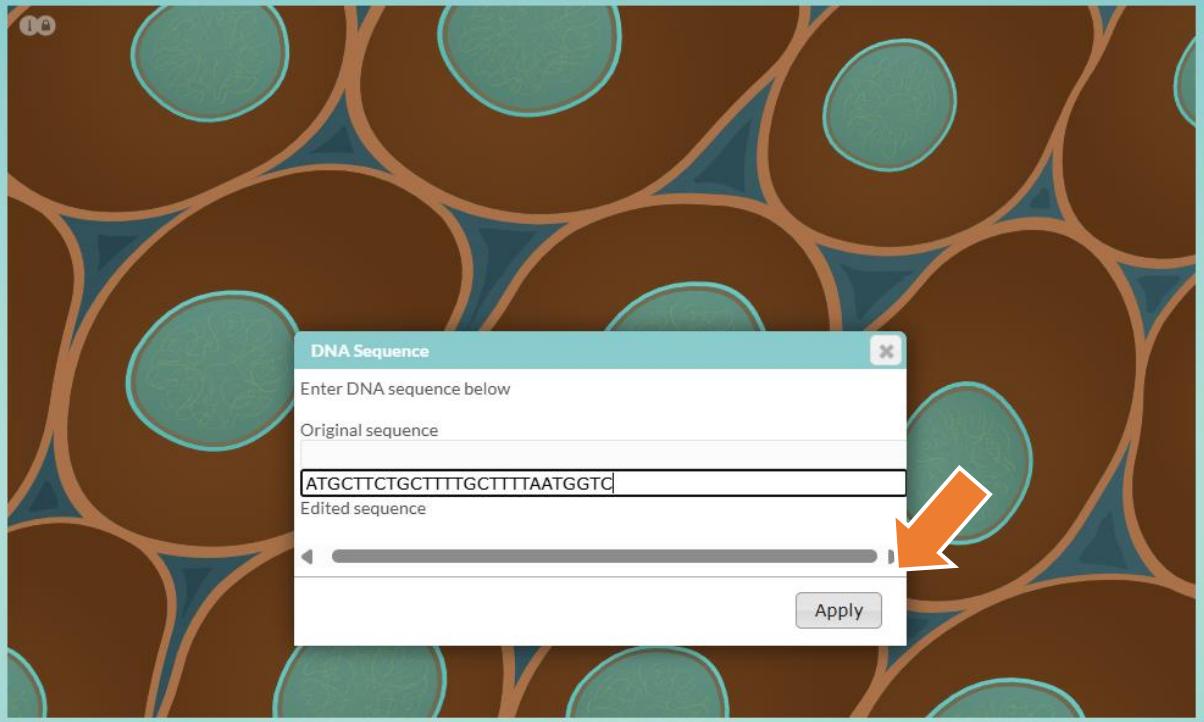
Step 1 : Enter or Edit a DNA Sequence

Step 2: Transcribe to make an mRNA strand, or Transcribe one base at a time

Step 3: Translate to generate a segment of the protein or Translate one amino acid at a time

Controls: Pause Resume Reset << Skip Skip >> Unfold protein Refold protein

Step 1: Enter the sequence of study -> Apply



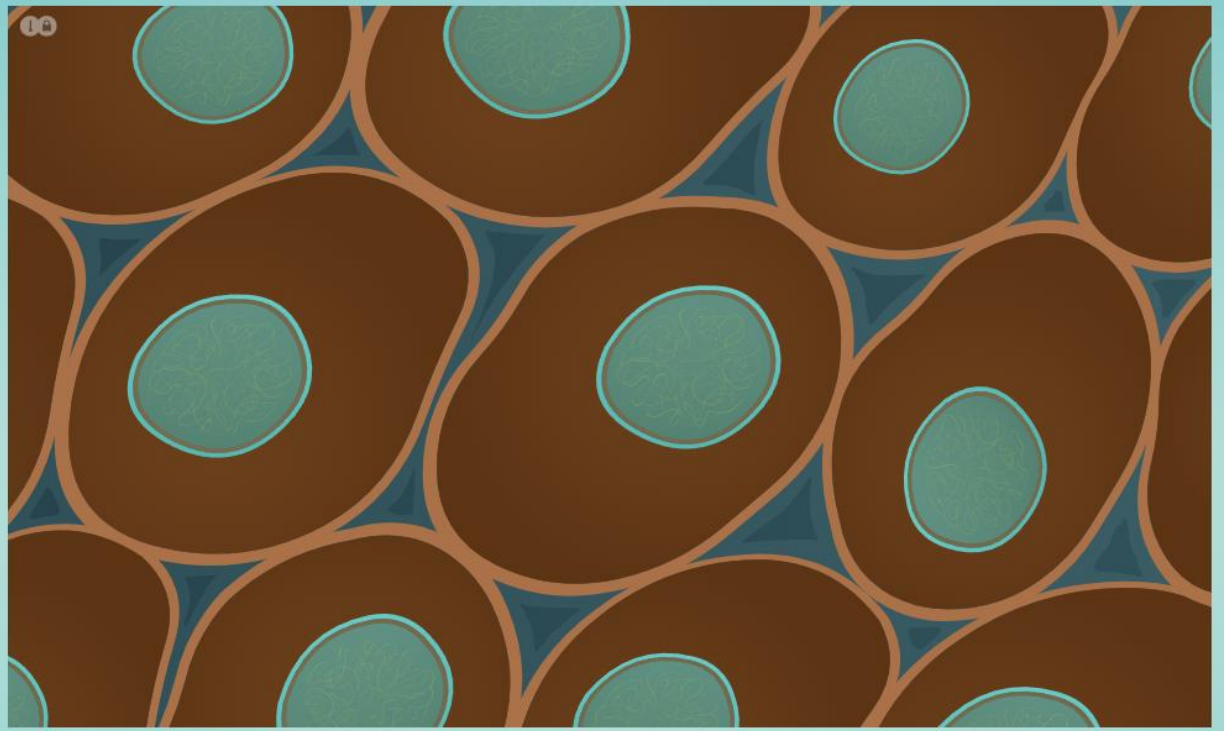
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Step 2: Transcribe the sequence



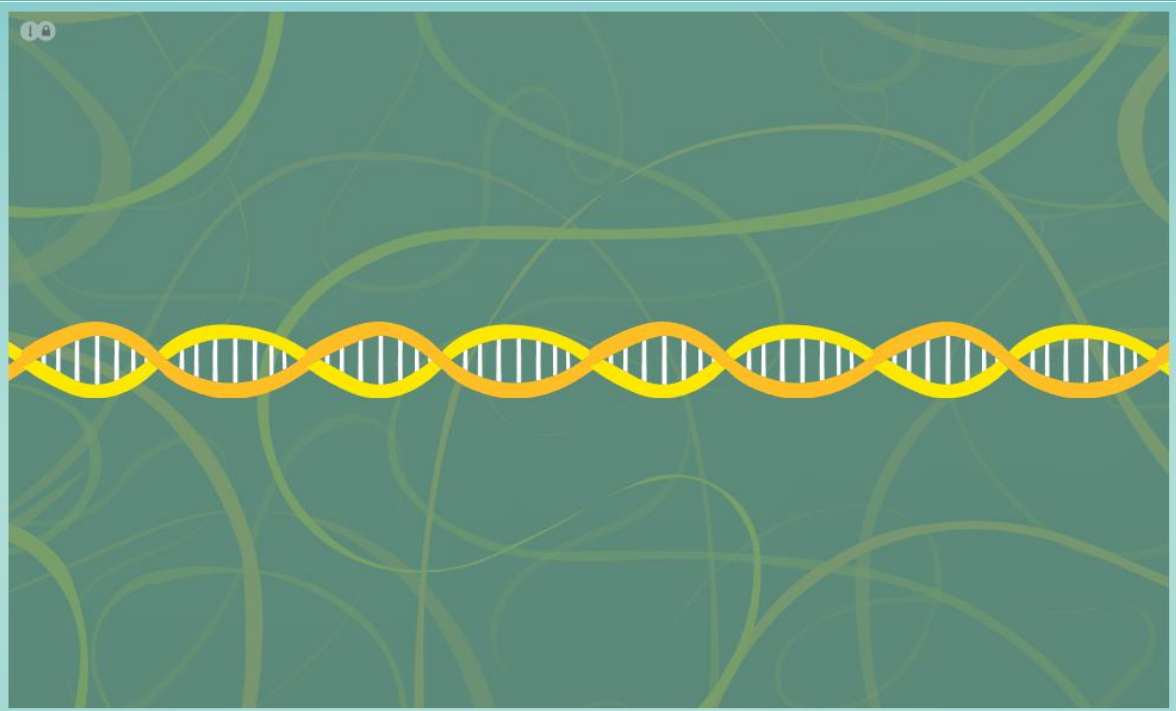
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Step 2: Transcribe the sequence



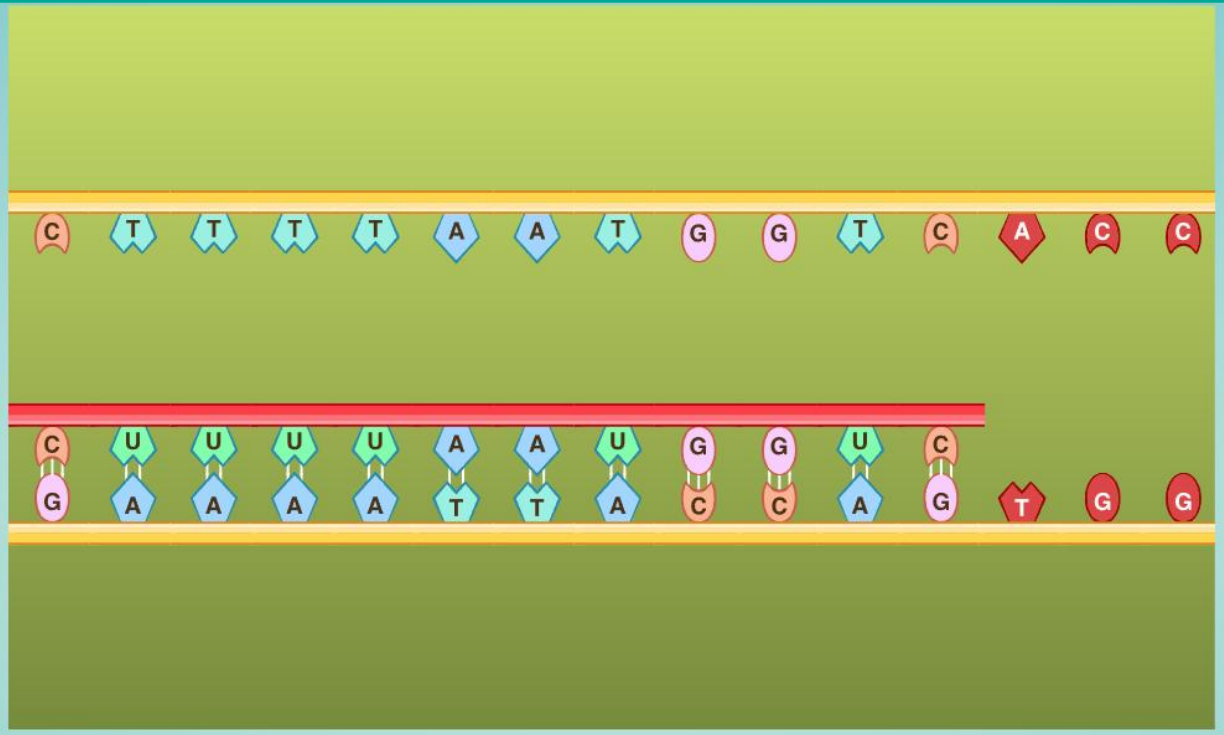
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Step 3: Translate the sequence



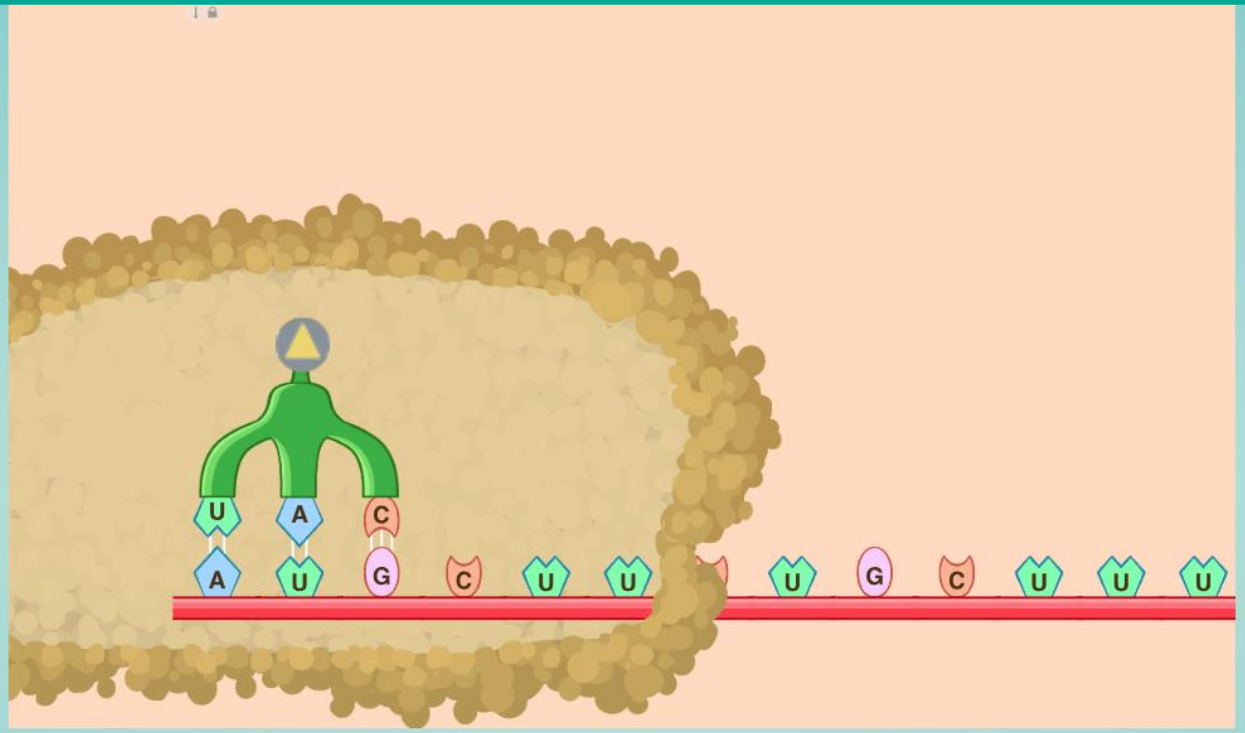
Step 1 : Enter or Edit a [illegible]quence

Step 2: Transcribe [illegible]make an mRNA strand, or Transcribe one base at a time

Step 3: Translate to generate a segment of the protein or Translate one amino acid at a time

Controls: Pause Resume Reset << Skip Skip >> Unfold protein Refold protein

Step 3: Translate the sequence



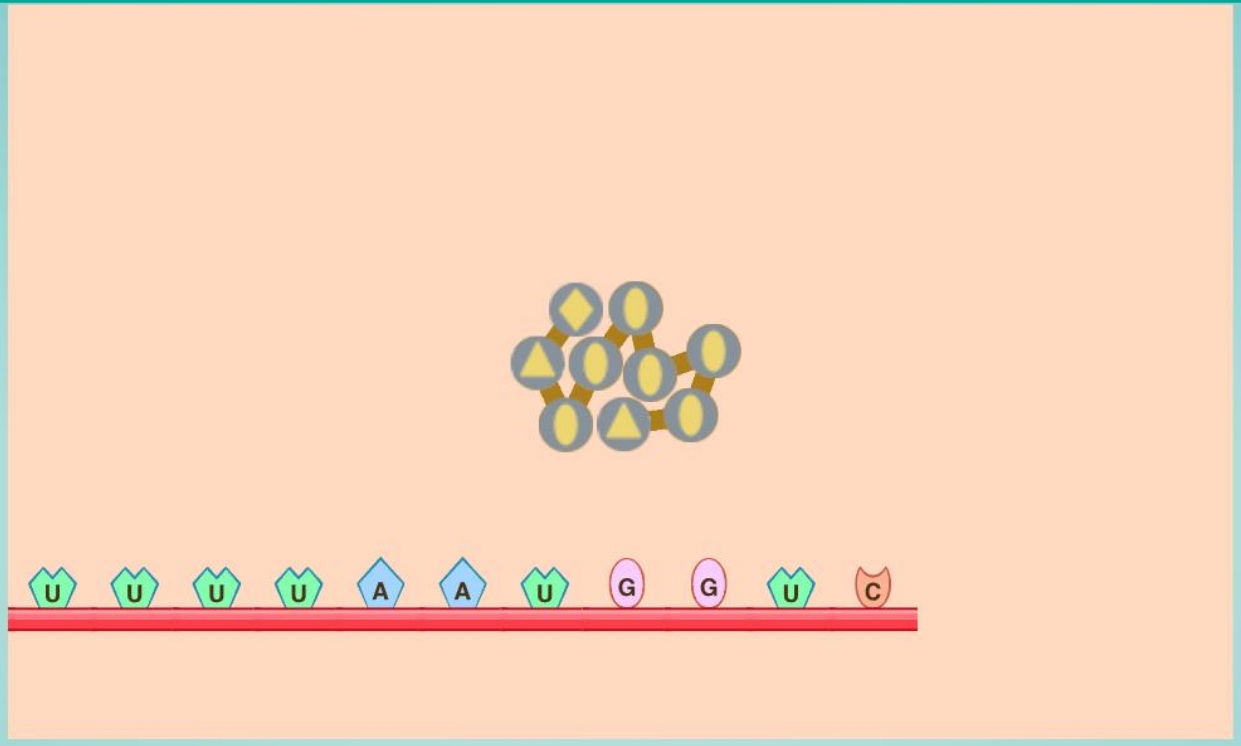
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Step 3: Translate the sequence



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Controls: Pause Resume Reset << Skip Skip >> Unfold protein Refold protein

Result

This 27-nucleotide sequence encodes **9 amino acids**:

Codon	Amino Acid
ATG	Met (M) – Start
CTT	Leu (L)
CTG	Leu (L)
CTT	Leu (L)
TTG	Leu (L)
CTT	Leu (L)
TTA	Leu (L)
ATG	Met (M)
GTC	Val (V)

Let’s add some polar amino acids…

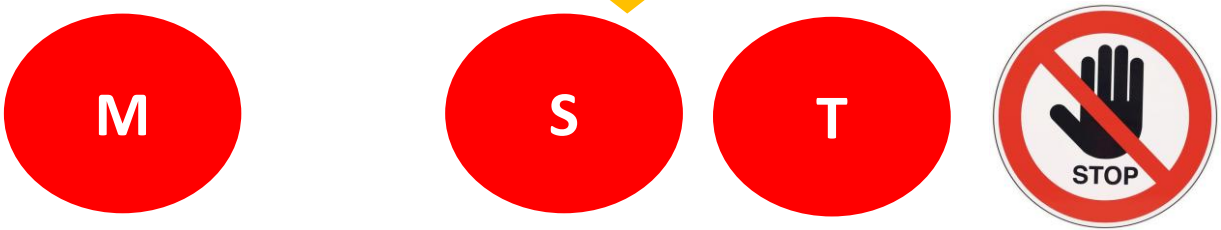
DNA

5' - ATG … AGT ACC TAA – 3'

RNA

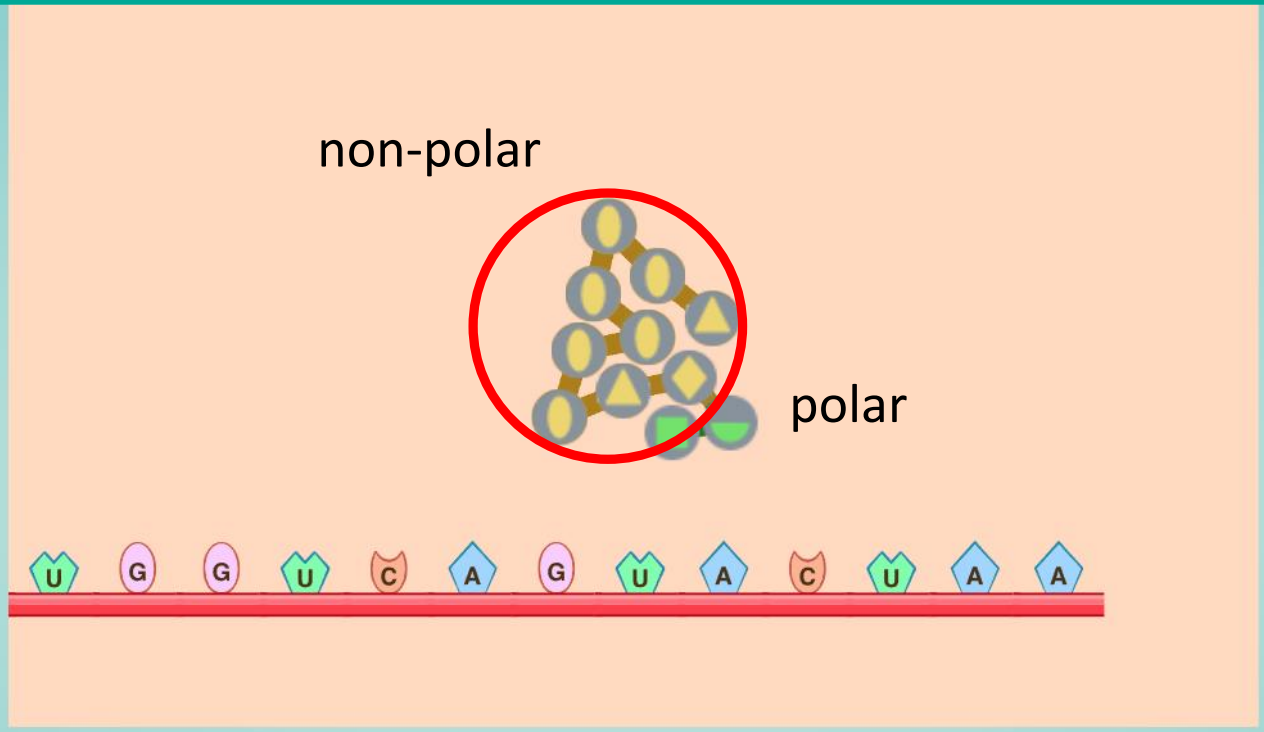
5' - AUG … ??? ??? UAA –3'

Protein



methionine serine threonine

Result



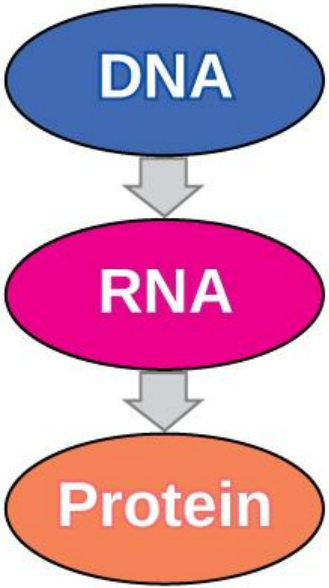
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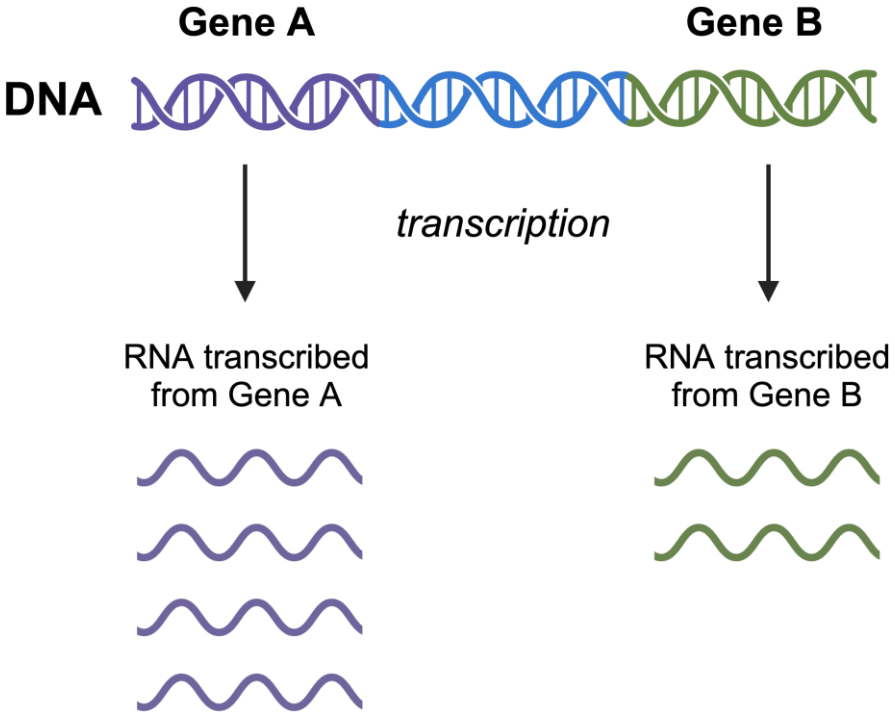
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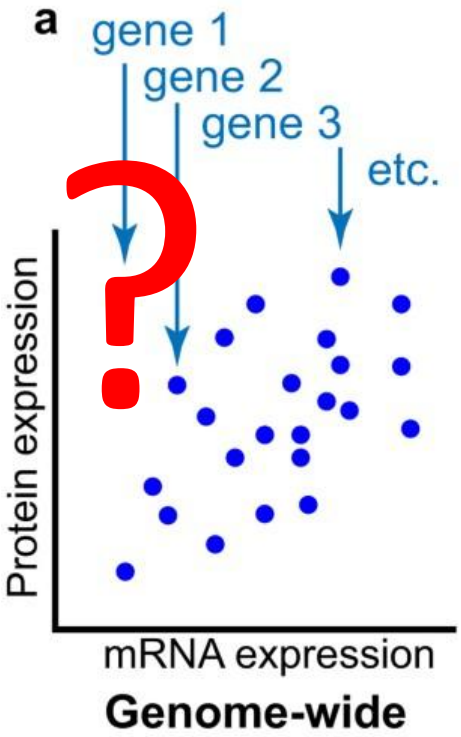
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Controls: Pause Resume Reset << Skip Skip >> Unfold protein Refold protein

How Genes Become Proteins — and Why Not All Genes Are Equal

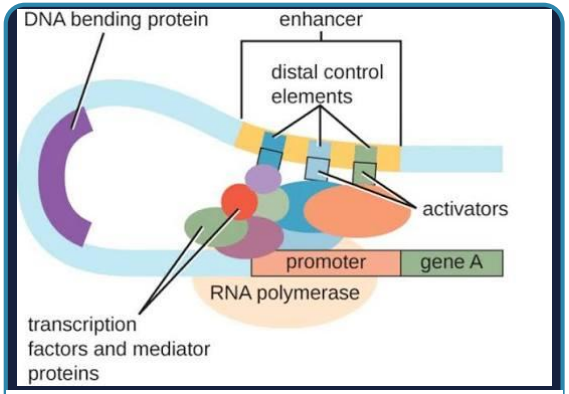






https://pressbooks.umn.edu/introbio/chapter/geneticstranscription/

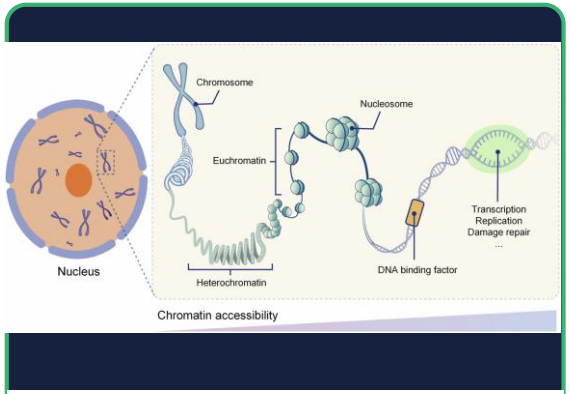
Why Are Not All Genes Transcribed the Same Way?



Transcription Factors

Proteins that bind to DNA and either activate or repress RNA polymerase.

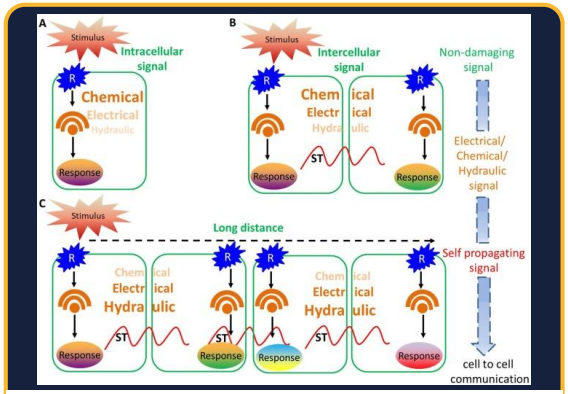
Example: In T-cells, NFAT activates *CTLA4* transcription only after antigen stimulation.



Chromatin Accessibility

DNA wrapped around histones may be open (euchromatin) or closed (heterochromatin).

Example: Methylation or acetylation of histones near *CTLA4* locus regulates its accessibility.



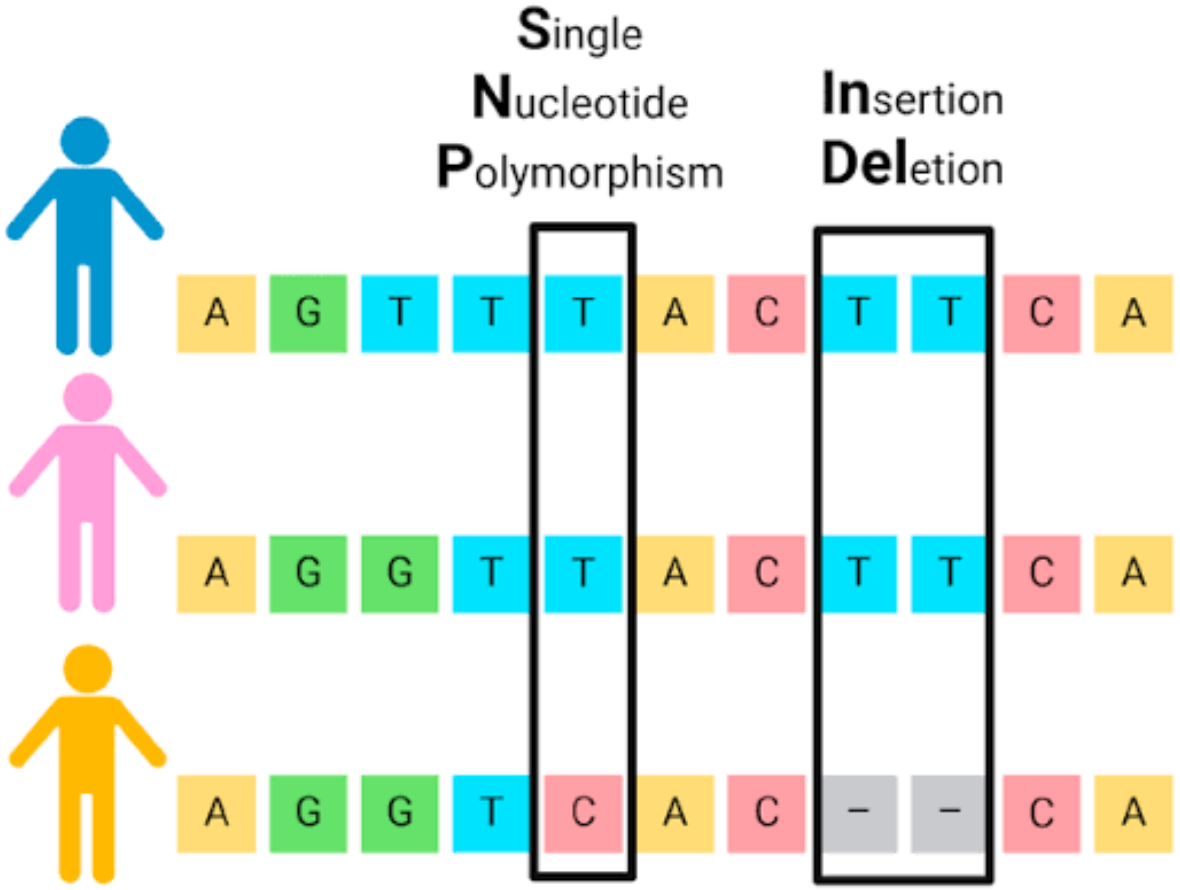
Cell Signaling Context

Signal cascades (e.g., TCR activation) trigger transcription only in specific cell types or conditions.

Example: *CTLA4* is expressed in activated T-cells but not in resting immune cells.

Gene expression is tightly regulated — the same DNA can produce very different outcomes depending on the cellular environment.

Polymorphisms and Mutations impacting the gene expression



CTLA4 Mutations That Impact Transcription

Promoter Region Mutations

SNP -318 C/T — Located in the *CTLA4* promoter. The T allele reduces transcription factor binding, lowering *CTLA4* expression and weakening immune checkpoints.

SNP -1661 A/G — Alters NF-kB binding site affinity, modifying transcription rate in activated T-cells.

3'UTR & Splicing Variants

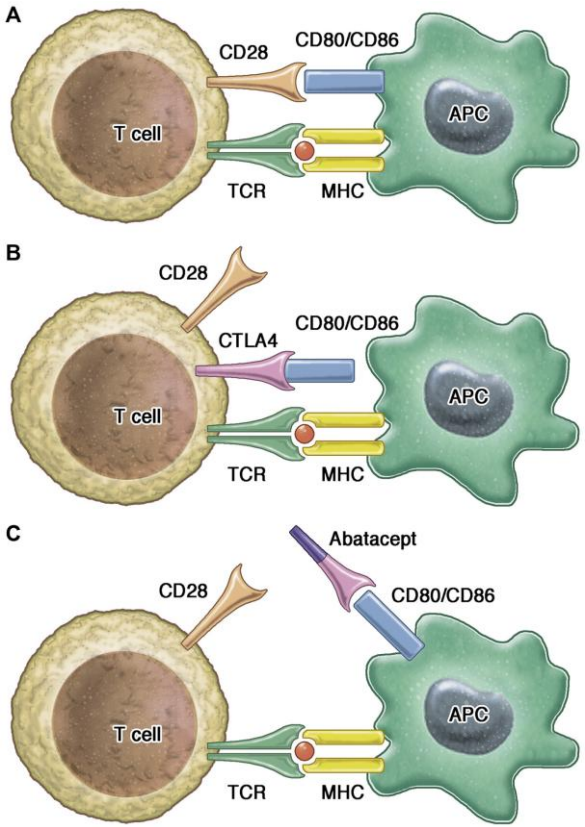
CT60 (rs3087243) A/G — In the 3'UTR, this SNP affects mRNA stability and alternative splicing, reducing levels of soluble CTLA4 (sCTLA4) isoform.

Joanna et al. (2023) linked CT60 G-allele to lower sCTLA4 levels, associated with autoimmune diseases.

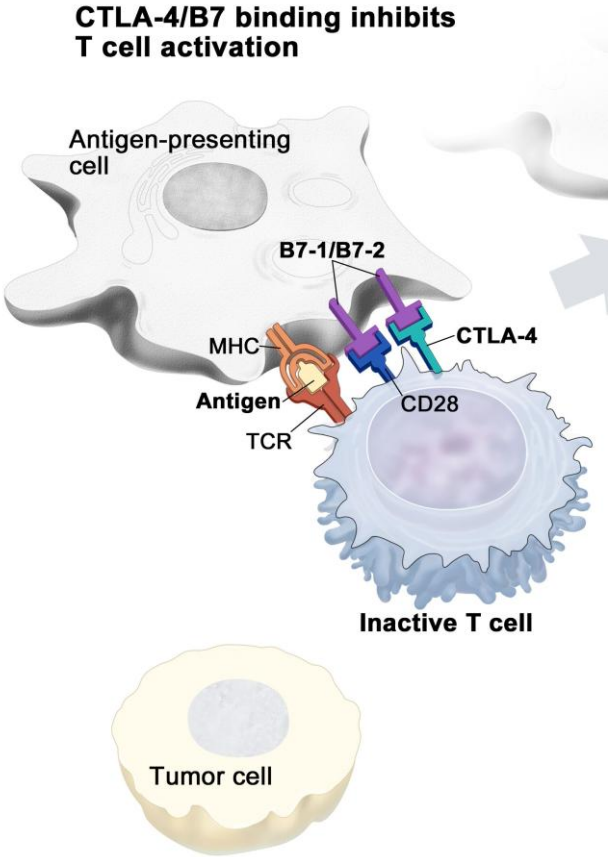
Clinical Impact of Reduced *CTLA4* Expression

Less CTLA4 → T-cells stay active longer → increased risk of autoimmune disease (Type 1 Diabetes, Rheumatoid Arthritis, Graves disease)
More CTLA4 → enhanced immune suppression → risk of tumor immune evasion — target of cancer immunotherapy (Ipilimumab blocks CTLA4)

Autoimmune diseases



Cancer Therapy Resistance





OMICS Sciences

The suffix '-omics' refers to large-scale analysis of classes of biomolecules. Multi-omics integrates these layers for a comprehensive view of biological systems.

The OMICS Landscape — Building on the Central Dogma

GenOMICS



DNA & genetic variation

Studies entire genome; SNPs, CNVs, structural variants. Foundation of precision medicine.

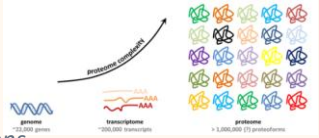
TranscriptOMICS



RNA expression

Quantifies mRNA and non-coding RNAs. Reveals gene activity in a specific cell state.

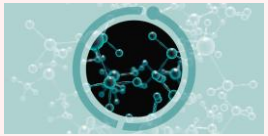
ProteOMICS



Proteins & interactions

Identifies and quantifies proteins; maps interaction networks. Closest to biological function.

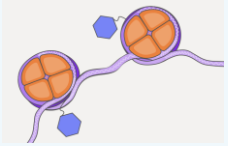
MetabolOMICS



Metabolites & pathways

Profiles small molecules. Reflects the phenotypic output of all upstream molecular events.

EpigenOMICS



Epigenetic modifications

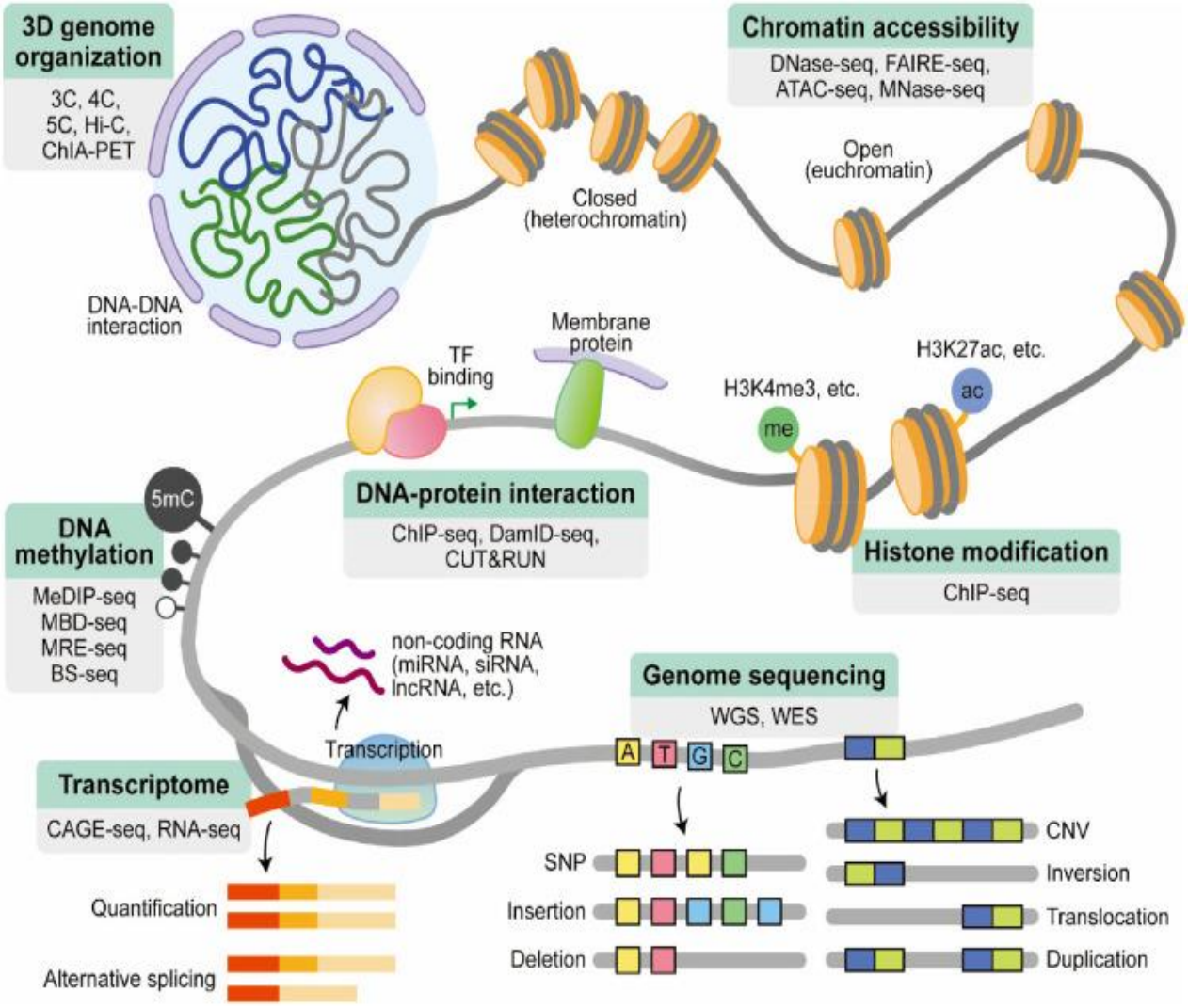
Studies DNA methylation, histone modifications. Explains heritable changes without altering DNA sequence.

LipidOMICS



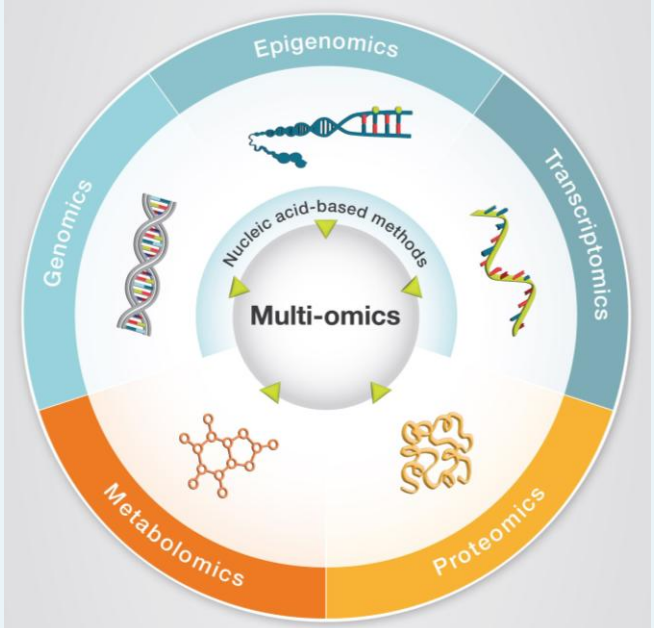
Lipid pathways & biomarkers

Characterises lipid species; important in cancer, cardiovascular and metabolic diseases.


3D genome organization
3C, 4C, 5C, Hi-C, ChIA-PET
DNA-DNA interaction
Chromatin accessibility
DNase-seq, FAIRE-seq, ATAC-seq, MNase-seq
Closed (heterochromatin)
Open (euchromatin)
TF binding
Membrane protein
H3K4me3, etc.
me
H3K27ac, etc.
ac
DNA-protein interaction
ChIP-seq, DamID-seq, CUT&RUN
Histone modification
ChIP-seq
5mC
DNA methylation
MeDIP-seq
MBD-seq
MRE-seq
BS-seq
non-coding RNA (miRNA, siRNA, lncRNA, etc.)
Transcription
Genome sequencing
WGS, WES
A T G C
Transcriptome
CAGE-seq, RNA-seq
Quantification
Alternative splicing
SNP
Insertion
Deletion
CNV
Inversion
Translocation
Duplication

Multi-OMICS: Integrative Systems Biology

- **Multi-omics combines data across multiple OMIC layers for a holistic molecular view**
- **Single-omics captures one layer** — multi-omics reveals molecular crosstalk invisible in isolation
- Enables discovery of biomarkers, disease mechanisms and drug targets
- Example — **Breast cancer subtypes: integrating genomics + transcriptomics + proteomics → identifies clinically distinct subtypes, informing therapy selection**
- Tools for integration: MOFA (Multi-Omics Factor Analysis), mixOmics, iCluster, DIABLO
- Challenge: **data heterogeneity, batch effects, high dimensionality — bioinformatics is essential**





Which OMICS to Choose?

Apply your OMICS knowledge to real clinical and research scenarios. Choose the appropriate technique and justify your choice.

⍰ Exercise A

Which OMICS Technique Would You Use? (Part 1)

CASE 1

A patient with colorectal cancer. Oncologist wants to test for a *KRAS* mutation to guide personalised therapy.

→ Which OMICS?

CASE 2

A new inhibitor reduces cell proliferation, but the company doesn't know which proteins are actually affected by the treatment.

→ Which OMICS?

CASE 3

A researcher studies tumour cells with a *BRCA1* mutation and wants to understand which genes are being activated or repressed.

→ Which OMICS?

Hint: Think about what biological layer each question targets: DNA sequence? Protein abundance? Gene expression?

⍰ Exercise B

Which OMICS Technique Would You Use? (Part 2)

CASE 1

In colorectal cancer cell lines, loss of *MLH1* expression is observed without any mutation in the DNA sequence.

→ Which OMICS?

CASE 2

Following treatment with a new anticancer drug, the lab hypothesised an accumulation of lactate and a reduction in ATP in the tumour cells.

→ Which OMICS?

CASE 3

Two patients receive the same anticoagulant dose, but one develops severe bleeding. Doctors want to know if this relates to a *CYP2C9* variant.

→ Which OMICS?

Hint: MLH1 is a gene. Lactate and ATP are small molecules. CYP2C9 is an enzyme encoded by a gene. What layer is affected?

✓ Case Study — Answer Key

Case 1 — KRAS mutation

GenOMICS

Looking for a specific DNA sequence variant

Case 2 — Proteins affected by inhibitor

ProteOMICS

Directly quantifies protein abundance & modifications

Case 3 — Genes activated/repressed by BRCA1

TranscriptOMICS

Measures mRNA expression (gene activation/repression)

Case 4 — MLH1 loss without DNA mutation

EpigenOMICS

DNA methylation silences gene expression without altering sequence

Case 5 — Lactate & ATP accumulation

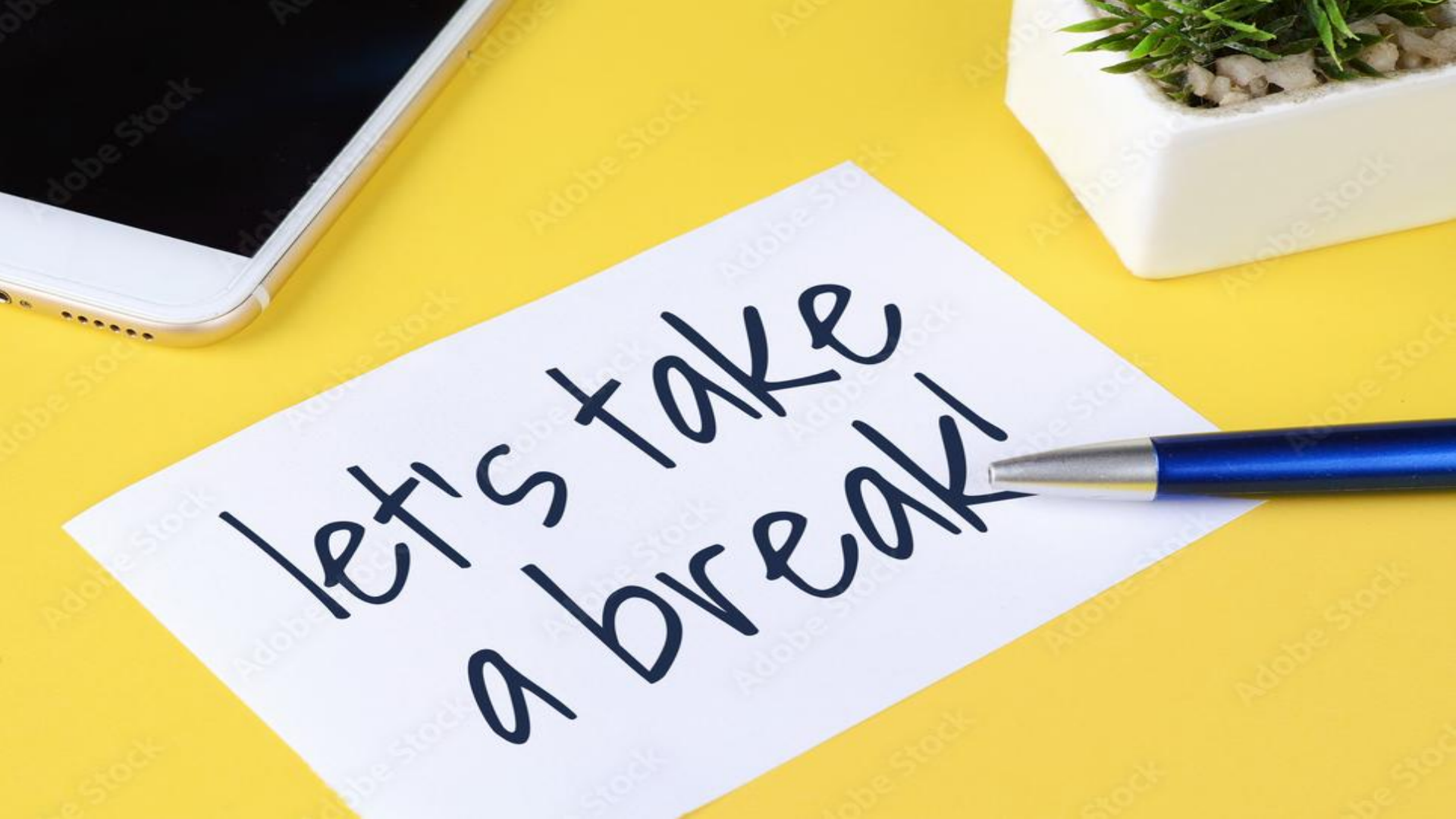
MetabolOMICS

Small metabolites (lactate, ATP) = metabolomics territory

Case 6 — CYP2C9 variant & drug metabolism

GenOMICS / Pharmacogenomics

DNA variant in a metabolising enzyme gene


let's take
a break!



Next-Generation Sequencing (NGS)

Massively parallel sequencing that delivers ultra-high throughput at a fraction of the cost of Sanger sequencing — revolutionising biology, medicine and drug discovery.

Sanger Sequencing → NGS: Why the Leap Mattered

Sanger Sequencing

- Frederick Sanger, 1977
- Based on chain-terminating dideoxynucleotides (ddNTPs)
- Generates fragments of different lengths terminated at each base
- **Separated by gel electrophoresis** → sequence read
- Gold standard for small targets; **still used for validation**
- Cost: ~$3 billion and 10+ years for one human genome
- Throughput: one sequence at a time

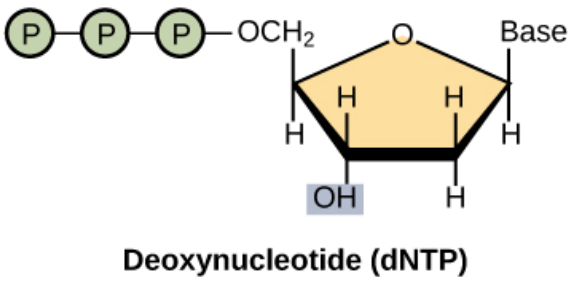
⚡ Next-Generation Sequencing (NGS)

- **Massively parallel — millions of reads simultaneously**
- Library prep: DNA fragmented → adapters ligated → clusters amplified
- **Multiple platforms: Illumina (short reads), Ion Torrent, PacBio, Oxford Nanopore**
- Illumina: bridge amplification on flow cell; fluorescent base detection
- Ion Torrent: pH change (H^+ release) detected electrically — no cameras
- Cost: <$200 per genome today; ~1–2 days runtime
- **Applications: WGS, WES, RNA-seq, ChIP-seq, metagenomics, ctDNA**

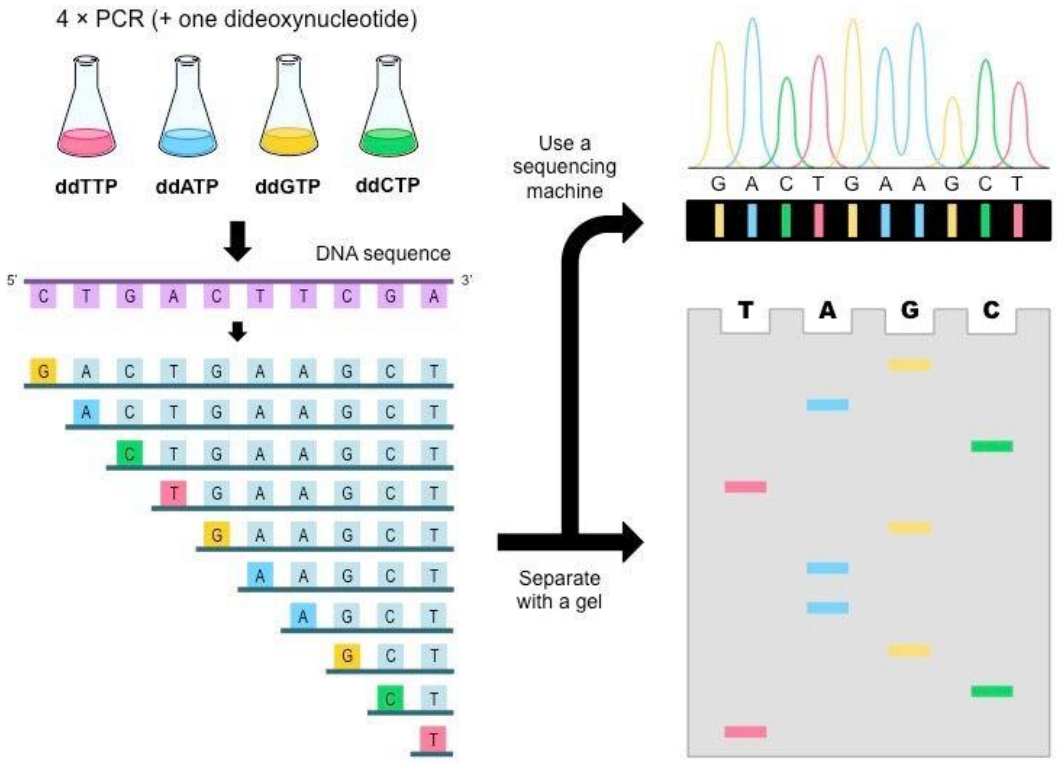
Sanger Sequencing → NGS: Why the Leap Mattered

Human Genome Project

- ***Sanger's Methodology***



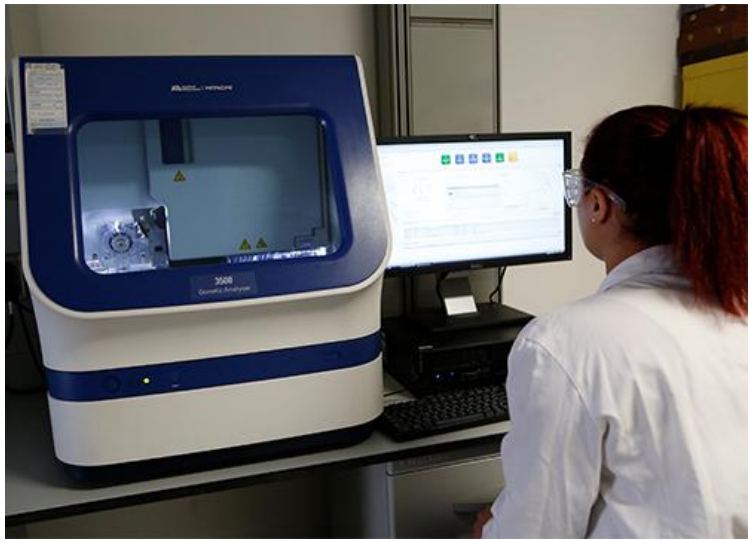
Deoxynucleotide (dNTP)

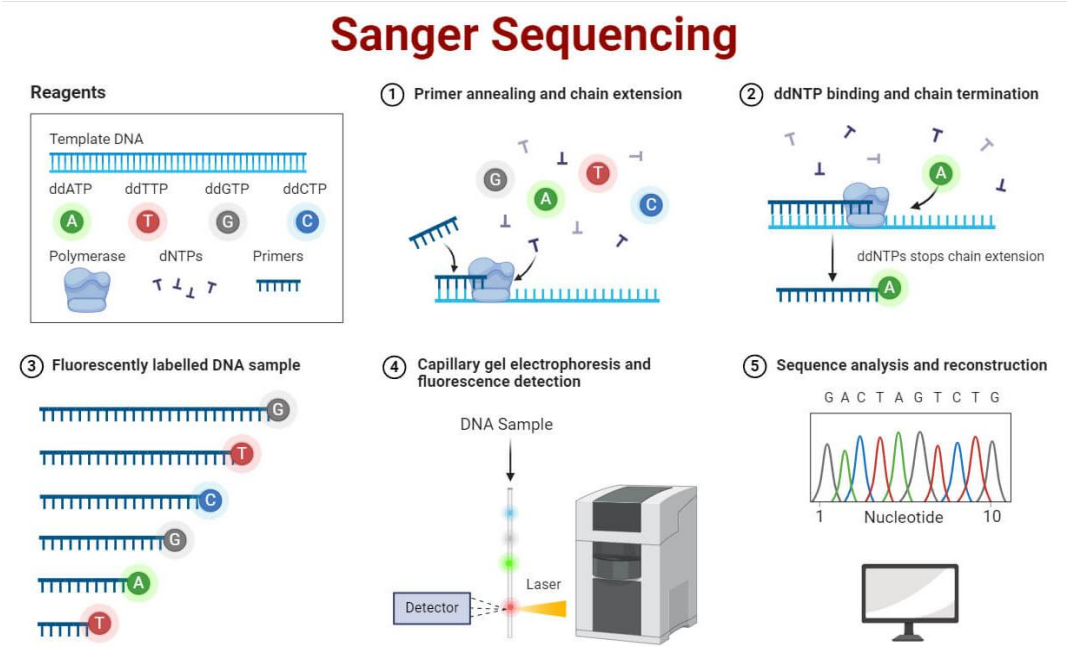


Sanger Sequencing → NGS: Why the Leap Mattered

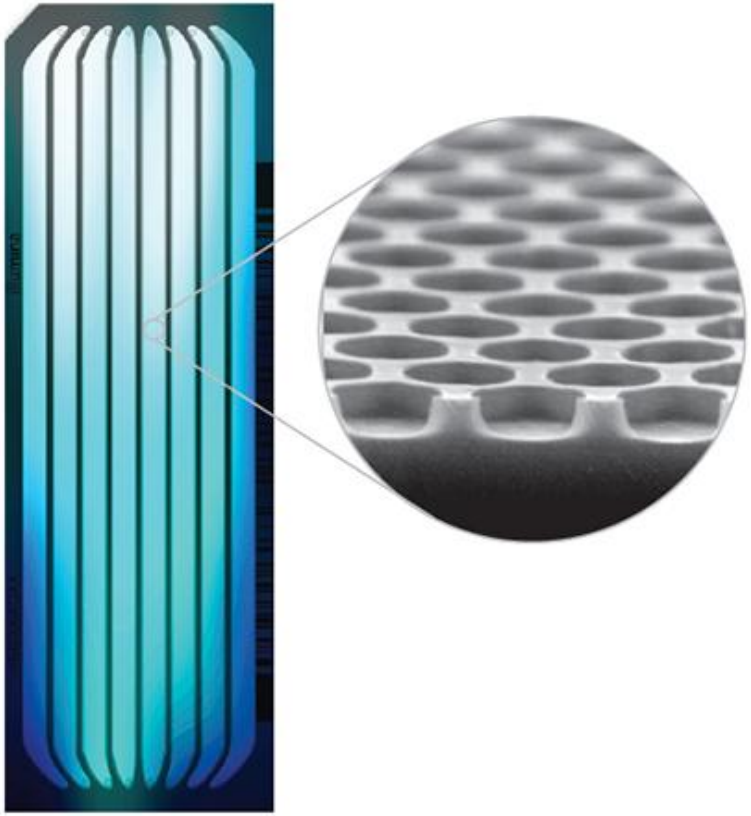
Human Genome Project

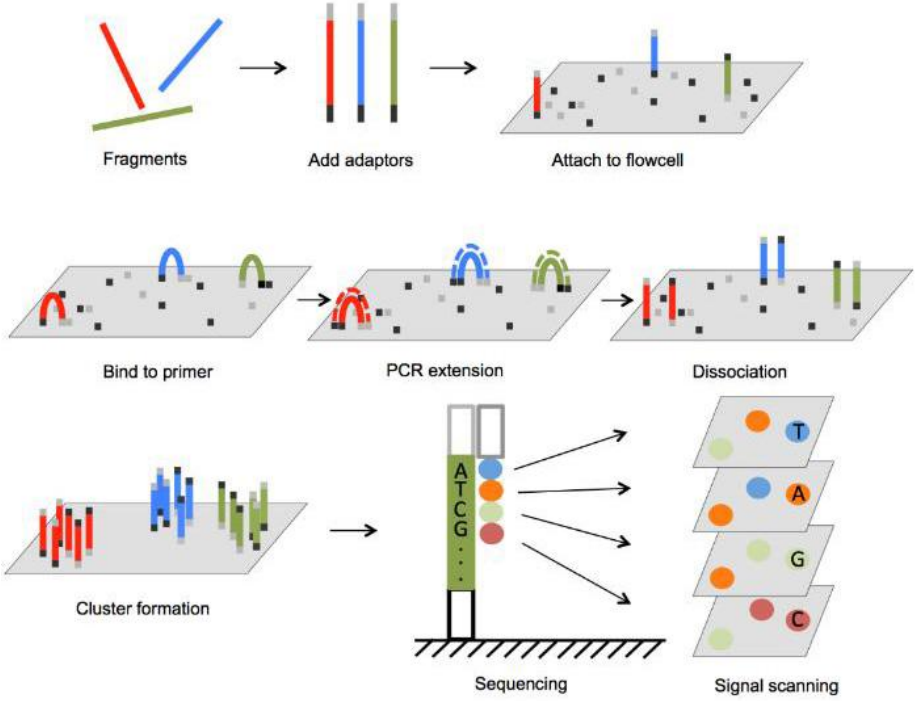
- ***Sanger's Methodology***



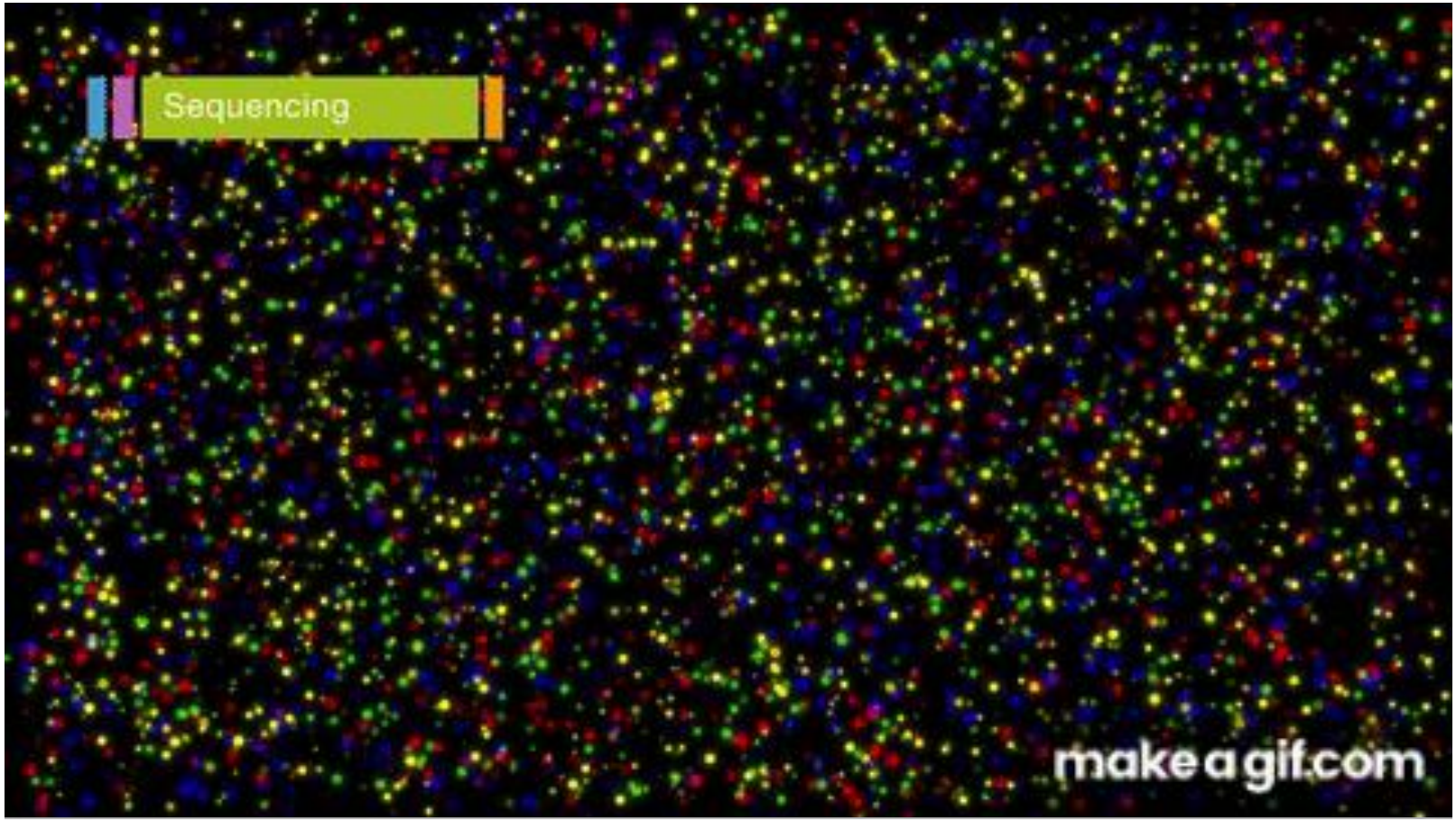


Next Generation Sequencing (NGS)





Next Generation Sequencing (NGS)

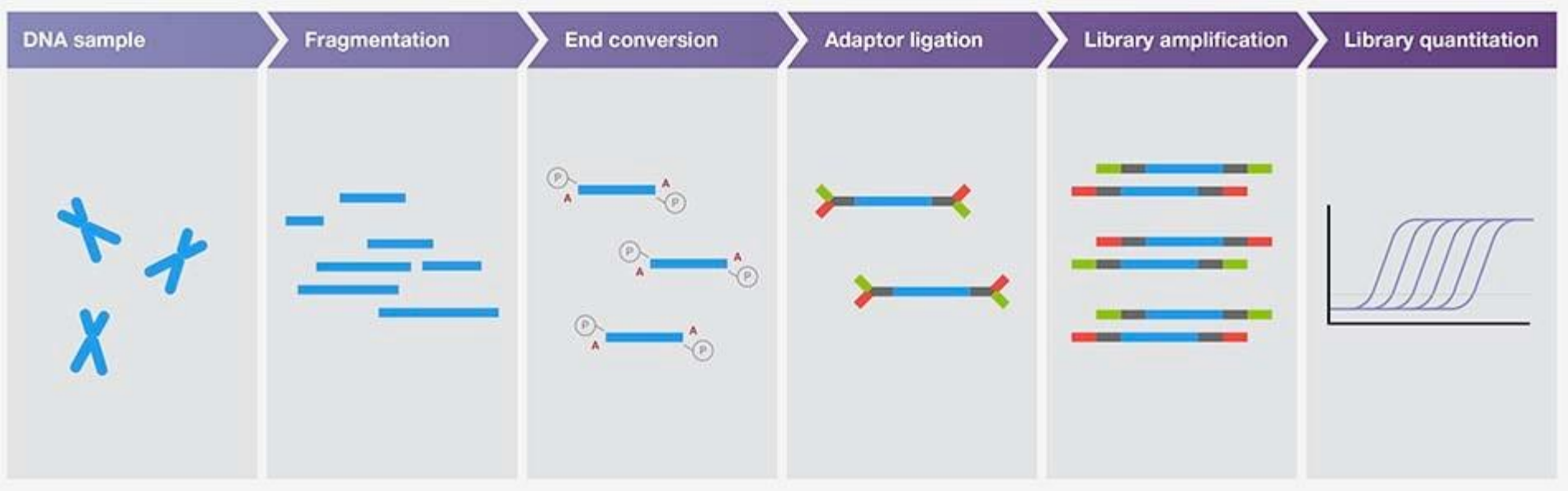


NGS Workflow: From Sample to Biological Insight



- Library prep: fragment DNA, ligate adapters, size-select, amplify
- Cluster amplification (Illumina): bridge amplification → millions of identical clusters on flow cell
- Sequencing by synthesis: fluorescently labelled reversible terminators; camera captures signal each cycle
- Bioinformatics pipeline: QC → alignment/assembly → variant calling → annotation → interpretation
- Output: FASTQ files → BAM/VCF → biological conclusions

Library Preparation



NGS Platforms: Illumina · Ion Torrent · PacBio · Oxford Nanopore

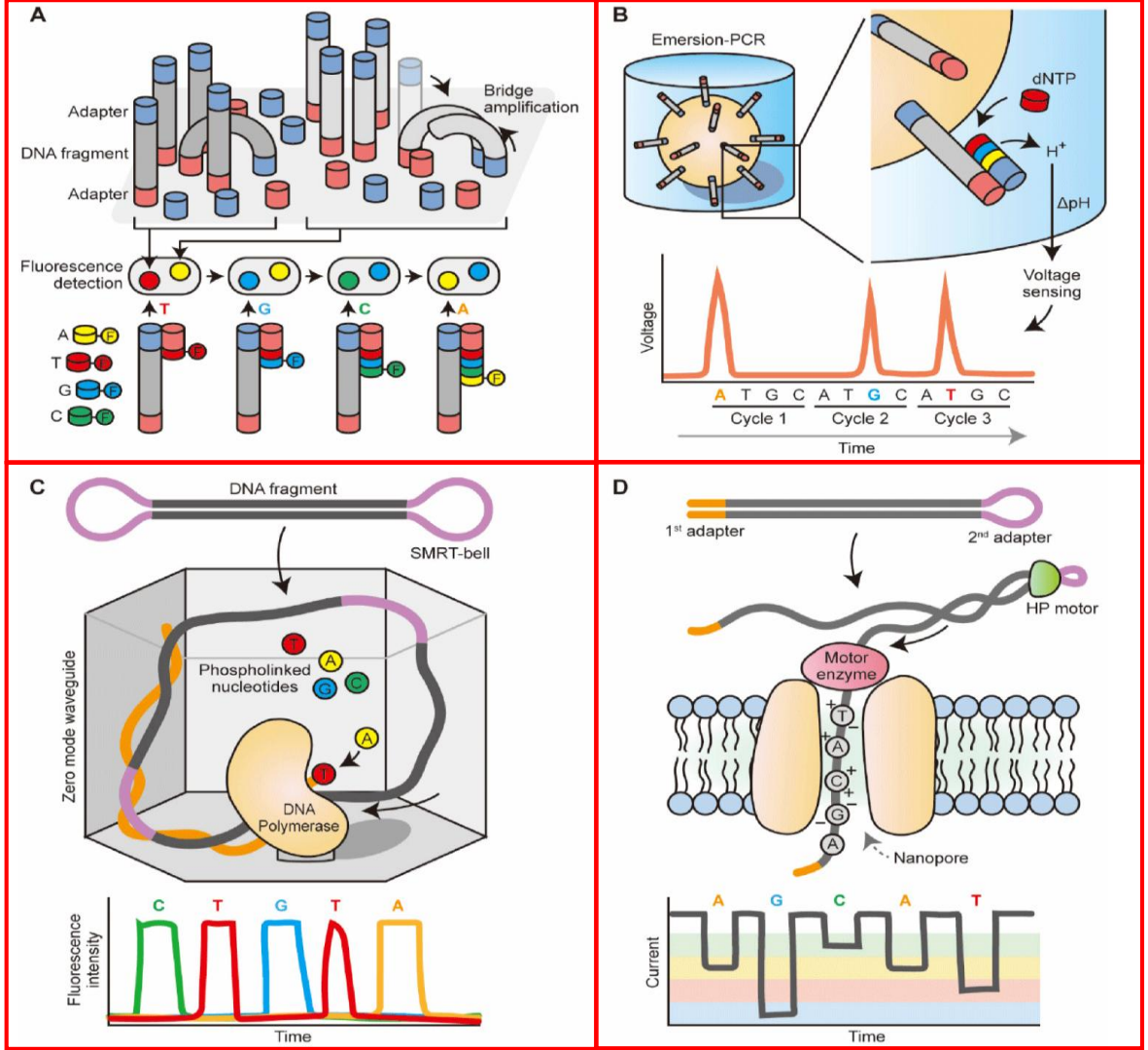
Platform	Generation	Read Length	Method	Error Rate	Best Use
Illumina	2nd	75–300 bp (short)	Sequencing by synthesis + bridge amplification	<0.1%	WGS, RNA-seq, ChIP-seq
⚡ Ion Torrent	2nd	200–600 bp	pH change (H^+ release) — semiconductor chip	~1% (homopolymers)	Targeted panels, clinical dx
PacBio SMRT	3rd	10–30 kb (long)	Single molecule real-time; fluorescent phospolabels	~0.1% (HiFi)	Genome assembly, structural variants
Oxford Nanopore	3rd	Up to Mb (ultra-long)	Electrical signal as DNA threads through nanopore	~5% raw (improving)	Real-time, field sequencing, full-length transcripts

illumina®

ion torrent
by Thermo Fisher Scientific

PacBio

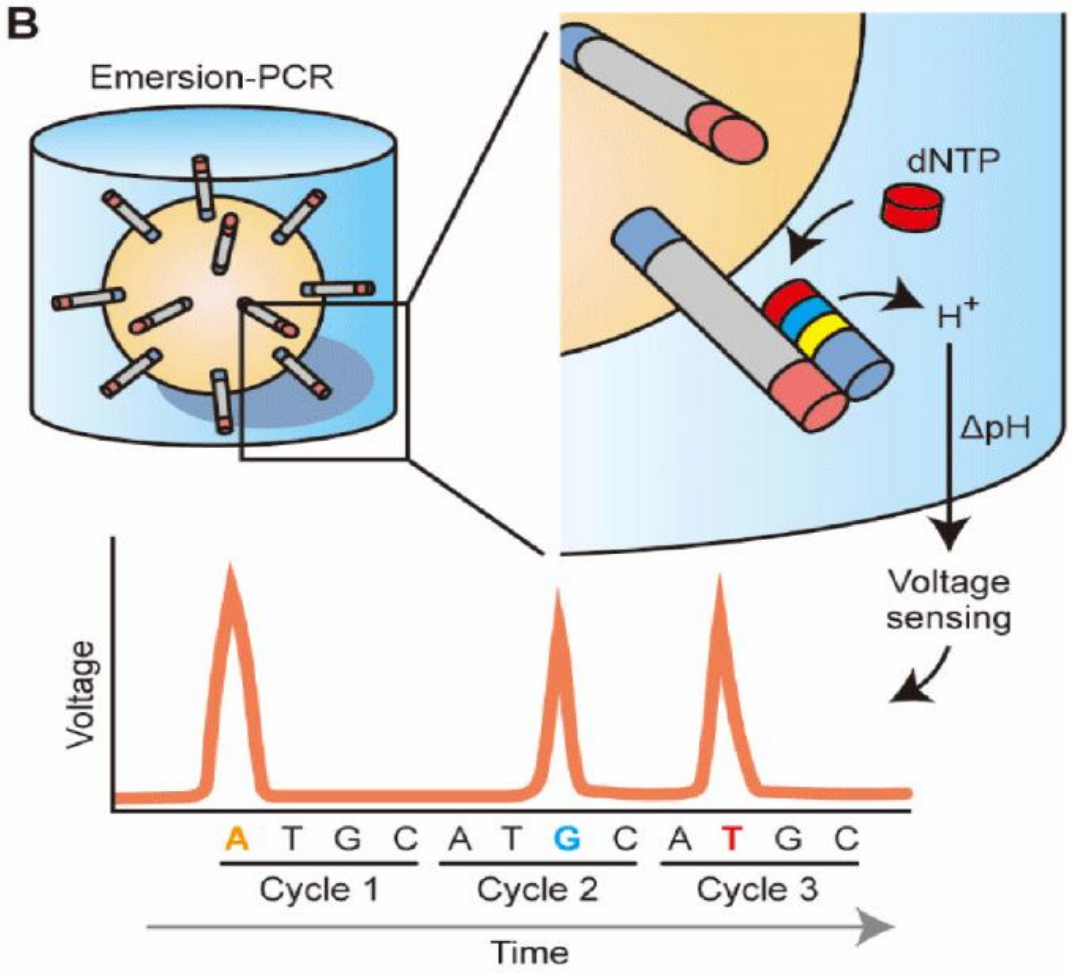
Oxford Nanopore Technologies



ion torrent

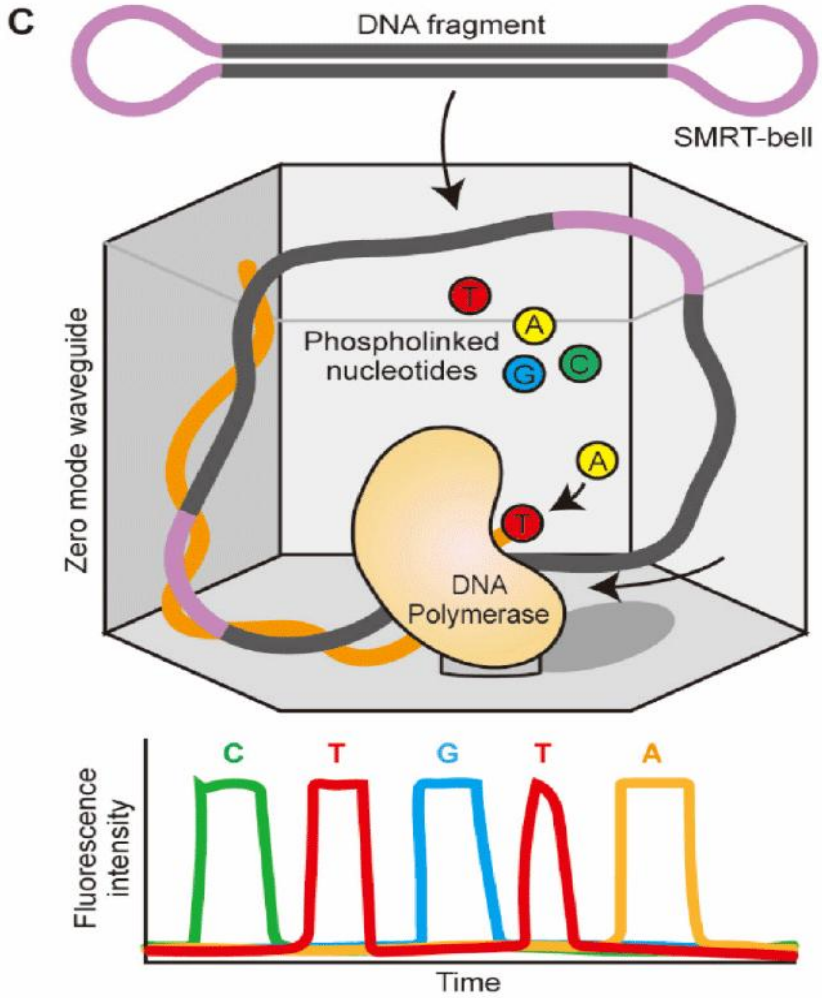
by **Thermo Fisher Scientific**





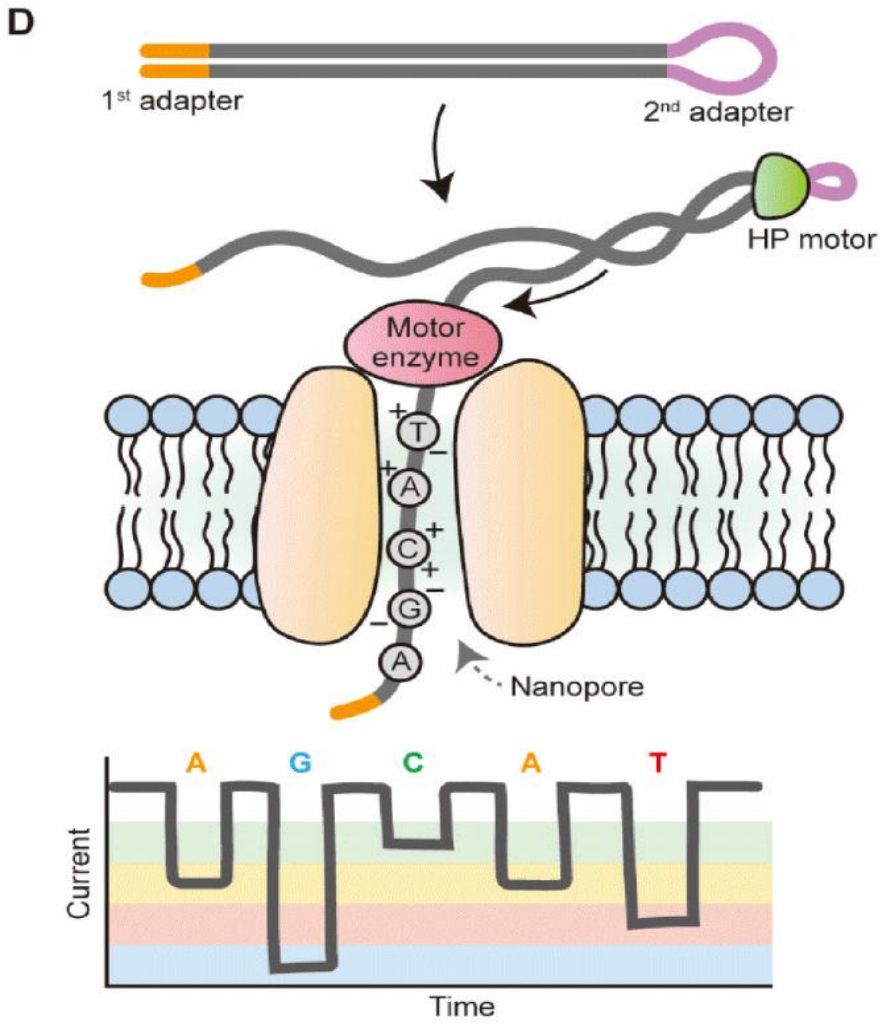
PacBio

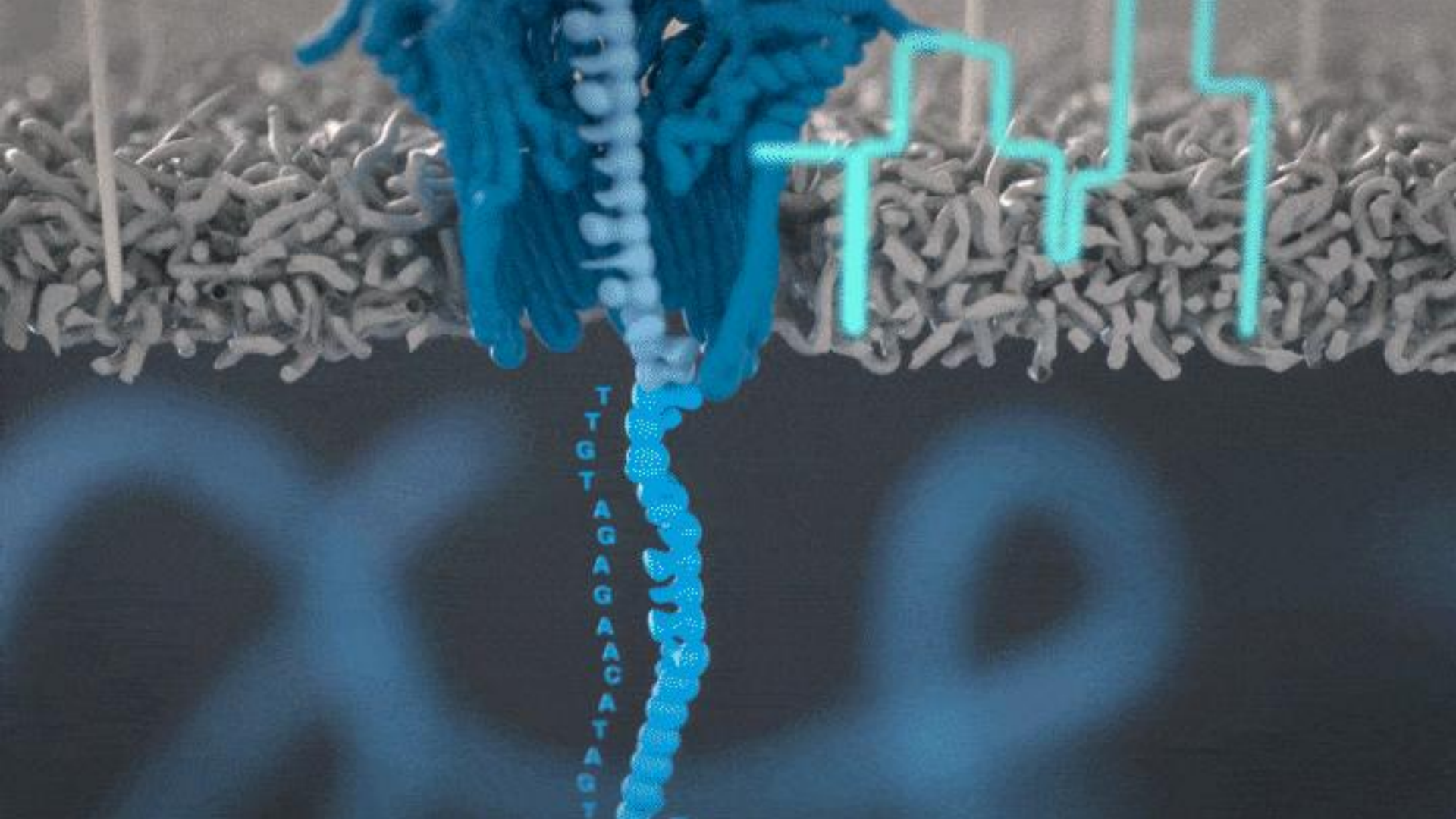



C
DNA fragment
SMRT-bell
Zero mode waveguide
Phospholinked nucleotides
T
A
G
C
A
T
DNA Polymerase
Fluorescence intensity
C
T
G
T
A
Time

Oxford Nanopore Technologies






TTGTAGAGAACATAGT

Genome NGS Applications Across Biology & Medicine

- Whole Genome Sequencing (WGS) — complete genome; rare disease diagnosis, cancer genomics
- Whole Exome Sequencing (WES) — ~2% of genome but ~85% of disease-causing variants
- Gene panels — targeted sequencing of clinically relevant genes (faster, cheaper for routine dx)
- ctDNA / liquid biopsy — detect tumour DNA in blood for non-invasive cancer monitoring



Part I — Key Takeaways

- Bioinformatics is the engine of modern biology — without computational tools, omics data is uninterpretable
- The Central Dogma (DNA → RNA → Protein) defines the layers that OMICS sciences analyse
- The Human Genome Project democratised genomics and revealed that complexity lies beyond gene count
- NGS has reduced sequencing costs by a million-fold and opened precision medicine at scale



R & RStudio— Setup

R is the statistical computing language of choice for bioinformatics.

R and RStudio — What and Why

R Programming Language

- Free, open-source statistical programming language (1996)
- **Standard in bioinformatics and biostatistics**
- Massive ecosystem: 20,000+ packages on CRAN and Bioconductor
- **Key bioinformatics packages: DESeq2, edgeR, limma, Seurat, ggplot2**
- **Bioconductor**: curated R packages specifically for genomic data analysis
- Used for: differential expression, survival analysis, GSEA, heatmaps, network analysis

RStudio IDE

- **Integrated Development Environment (IDE) for R — makes R accessible**
- Four panes: Console, Script editor, Environment, Files/Plots
- RStudio must have R installed first (R is the engine; RStudio is the dashboard)
- Free Desktop version available for Windows, macOS and Linux
- RStudio Cloud: browser-based — no installation needed for beginners
- Keyboard shortcuts: Ctrl+Enter to run, Ctrl+Shift+M for the pipe operator (%>%)

Installing R and RStudio — Step by Step

1

Download R

Go to cran.r-project.org → 'Download R for Windows' (or macOS) → click 'base' subdirectory → download installer (.exe or .pkg)

2

Install R

Run the installer, accept all default settings. R installs the base system and a basic graphical interface (RGui). Verify: open RGui and type version() to confirm.

3

Download RStudio

Go to posit.co/download/rstudio-desktop/ → 'Download RStudio for Windows/macOS' → download free version

4

Install RStudio

Run the installer. Open RStudio — it auto-detects R. Verify the console shows the R version. If not found, set it manually in Tools → Global Options → R General.

5

Install key packages

In the RStudio Console type: install.packages(c('BiocManager','ggplot2','dplyr'))
Then: BiocManager::install('DESeq2') for differential expression analysis.

Installing R and RStudio — Step by Step

- **Step 1:**
  - Vai al sito Comprehensive R Archive Network (CRAN) e clicca su “Download R for Windows” (or “Download R for MacOS”). **https://cran.r-project.org**
- **Step 2:**
  - Clicca su collegamento sottodirectory “base”.
- **Step 3:**
  - Clicca su “Download R-4.4.2 for Windows”. Il collegamento consente di scaricare un'estensione di installazione (.exe file).
- **Step 4:**
  - Eseguire il file .exe e seguire la procedura guidata di installazione accettando le impostazioni predefinite.

Una volta installato R, puoi procedere all'installazione di RStudio.

- **Step 5:**
  - Vai al sito Web di download di RStudio e fai clic su tasto “Download RStudio for Windows” (or the link for the MacOS version).
- **Step 6:**
  - Eseguire il file .exe e seguire le istruzioni di installazione.

Installing R and RStudio — Step by Step



CRAN
Mirrors
What's new?
Search
CRAN Team

About R
R Homepage
The R Journal

Software
R Sources
R Binaries
Packages
Task Views
Other

Documentation
Manuals

Download R-4.5.3 for Windows (87 megabytes, 64 bit)

README on the Windows binary distribution
New features in this version

This build requires UCRT, which is part of Windows since Windows 10 and Windows Server 2016. On older systems, UCRT has to be installed manually from here.

If you want to double-check that the package you have downloaded matches the package distributed by CRAN, you can compare the md5sum of the .exe to the fingerprint on the master server.

Frequently asked questions

- Does R run under my version of Windows?
- How do I update packages in my previous version of R?

Please see the R FAQ for general information about R and the R Windows FAQ for Windows-specific information.

Other builds

- Patches to this release are incorporated in the r-patched snapshot build.
- A build of the development version (which will eventually become the next major release of R) is available in the r-

Installing RStudio — Step by Step



creators of RStudio, built for the next generation of data science workflows with R and Python.

1: Install R

RStudio requires R 3.6.0+. Choose a version of R that matches your computer's operating system.

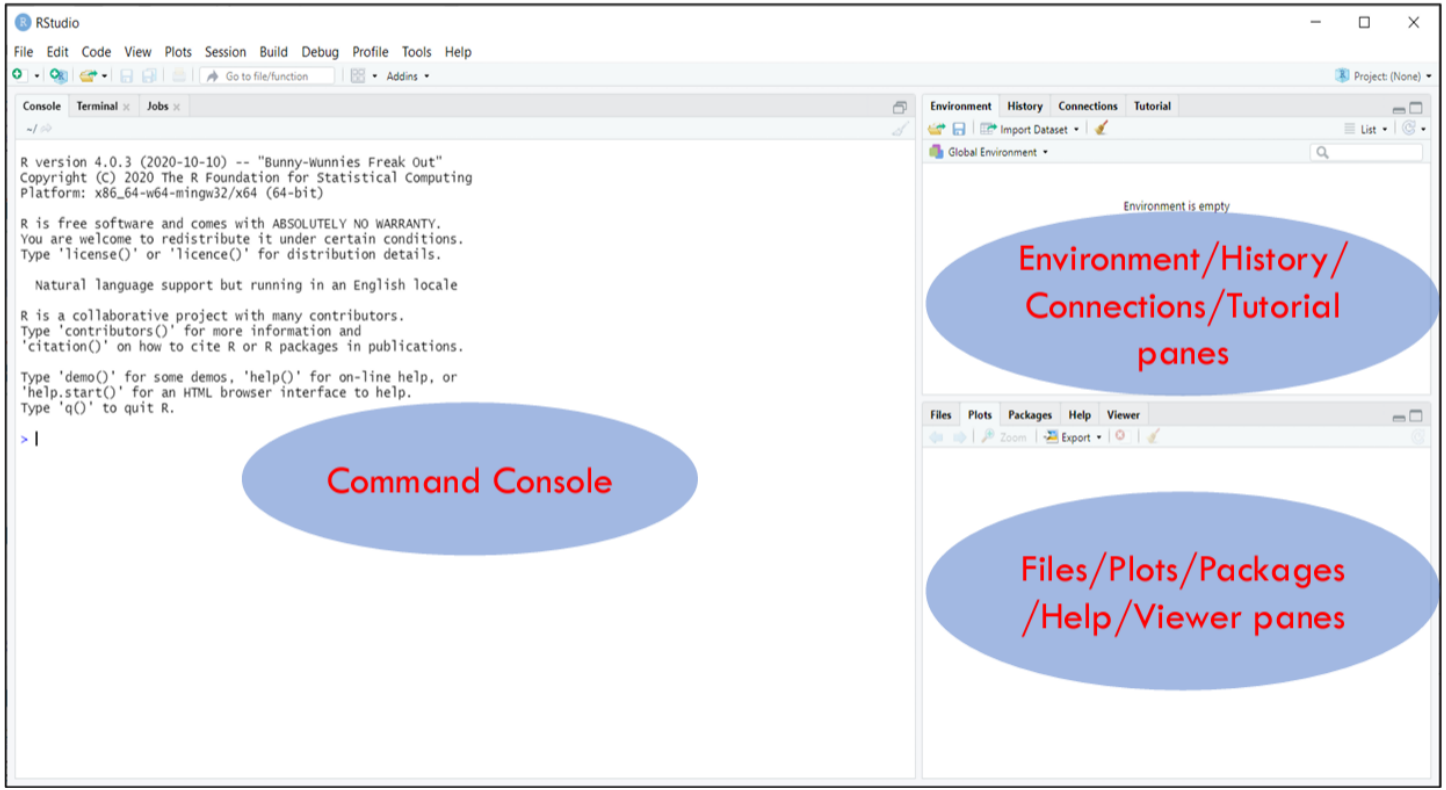
R is not a Posit product. By clicking on the link below to download and install R, you are leaving the Posit website. Posit disclaims any obligations and all liability with respect to R and the R website.

DOWNLOAD AND INSTALL R

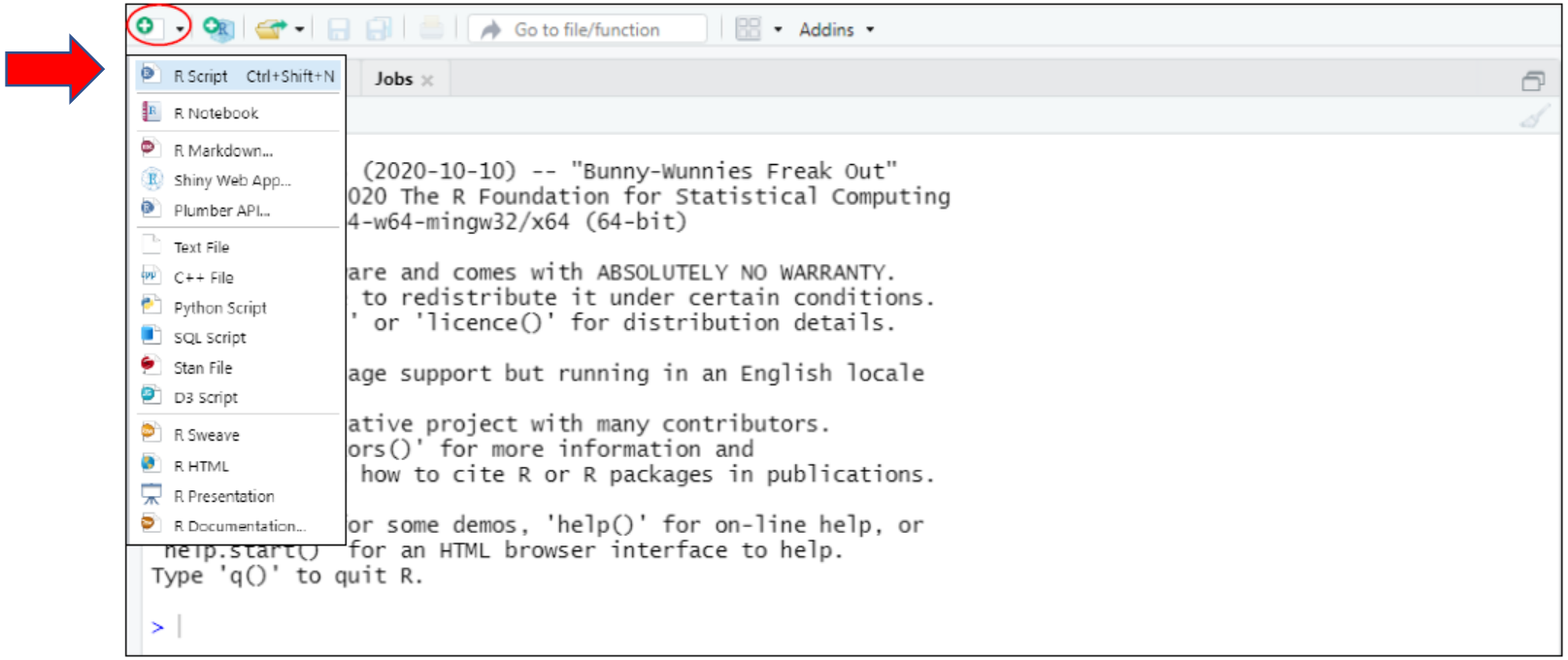
2: Install RStudio

DOWNLOAD RSTUDIO DESKTOP FOR WINDOWS

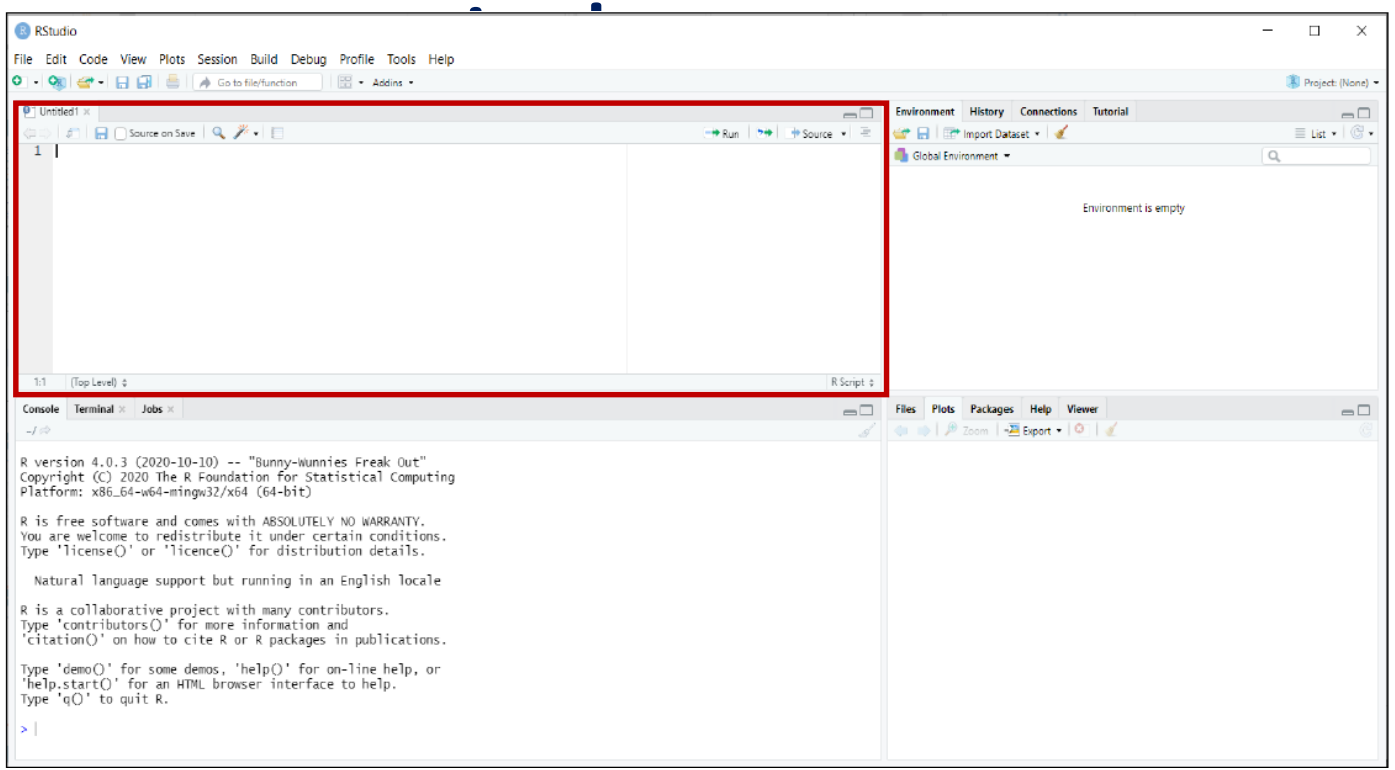
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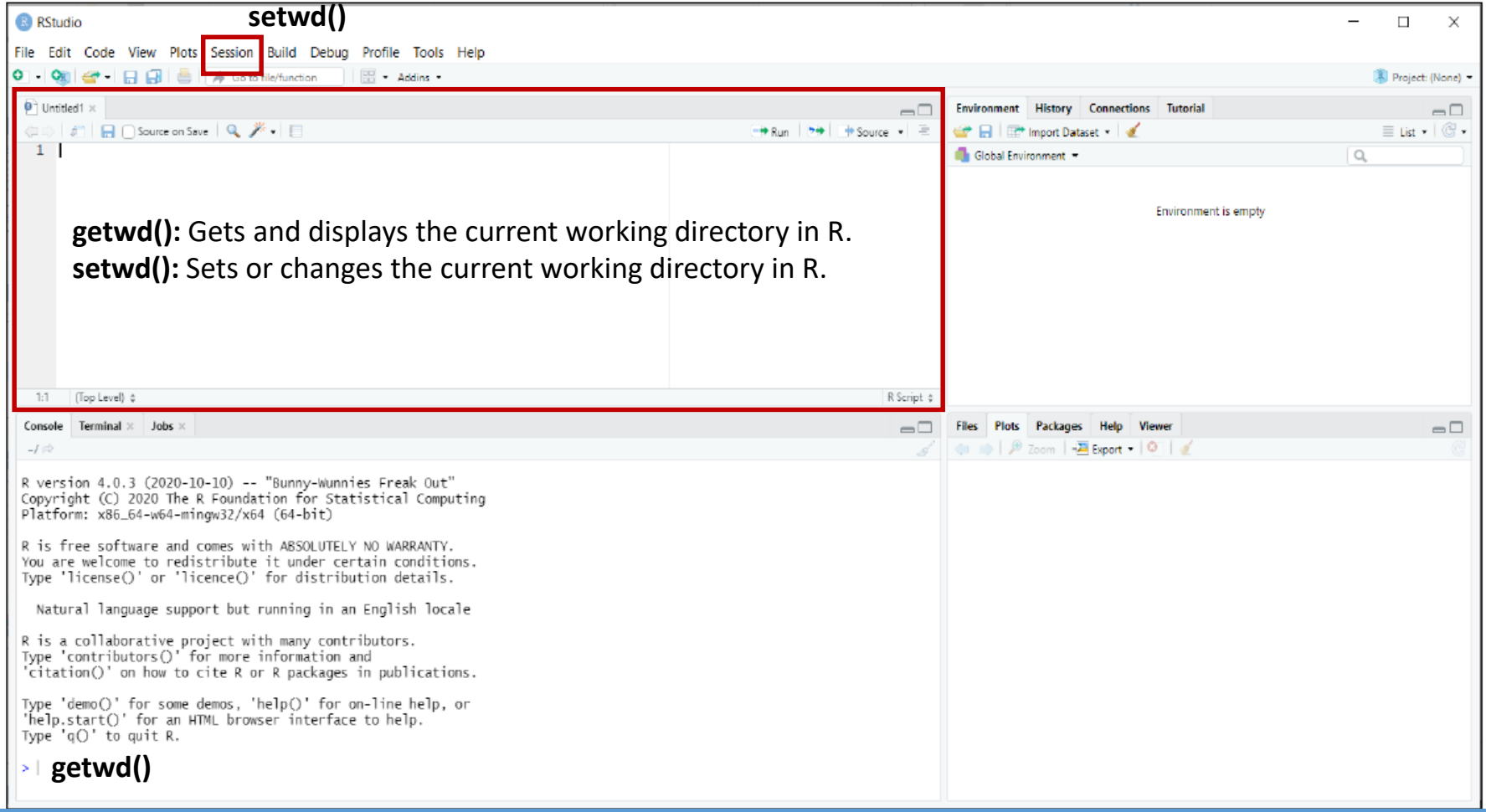

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Project: (None)
Console
Terminal
Jobs
R version 4.0.3 (2020-10-10) -- "Bunny-Wunnies Freak Out"
Copyright (C) 2020 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)
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'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
>
Command Console
Environment
History
Connections
Tutorial
Import Dataset
List
Global Environment
Environment is empty
Environment/History/
Connections/Tutorial
panes
Files
Plots
Packages
Help
Viewer
Zoom
Export
Files/Plots/Packages
/Help/Viewer panes

Rstudio: command console



Rstudio interface with editor




setwd()
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Addins
Project: (None)
Untitled1
Source on Save
Run
Source
1
getwd(): Gets and displays the current working directory in R.
setwd(): Sets or changes the current working directory in R.
1:1
(Top Level)
R Script
Environment History Connections Tutorial
Import Dataset
List
Global Environment
Environment is empty
Console Terminal Jobs
R version 4.0.3 (2020-10-10) -- "Bunny-Wunnies Freak Out"
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> getwd()
Files Plots Packages Help Viewer
Zoom
Export